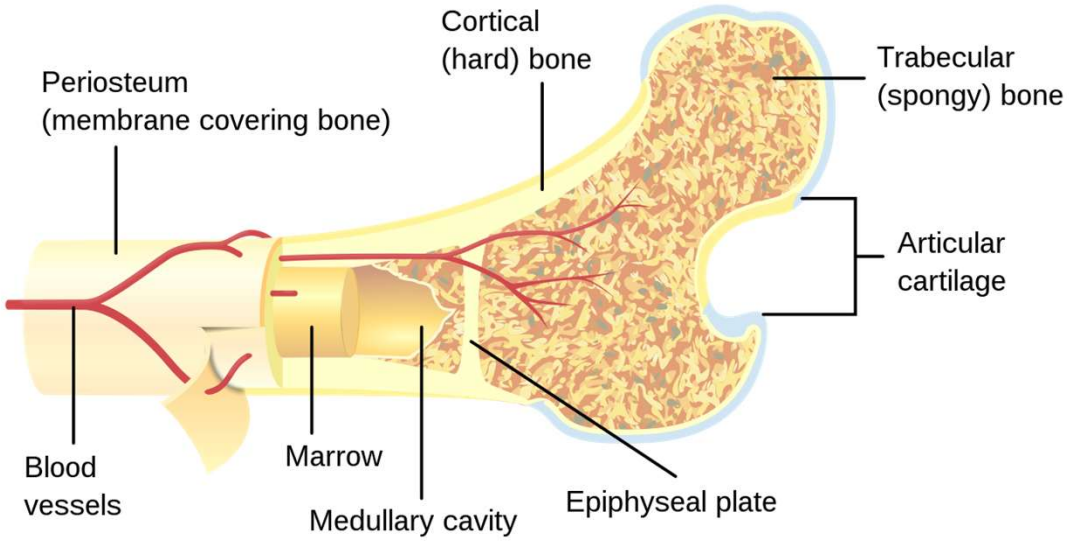
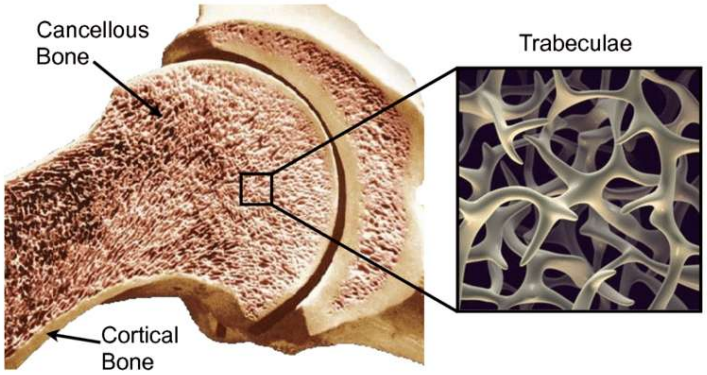
Biomedical Materials:
Ortheopedic implants (Hip Prosthesis)

Asst. Prof. Dr. Sumama

Introduction and Recap

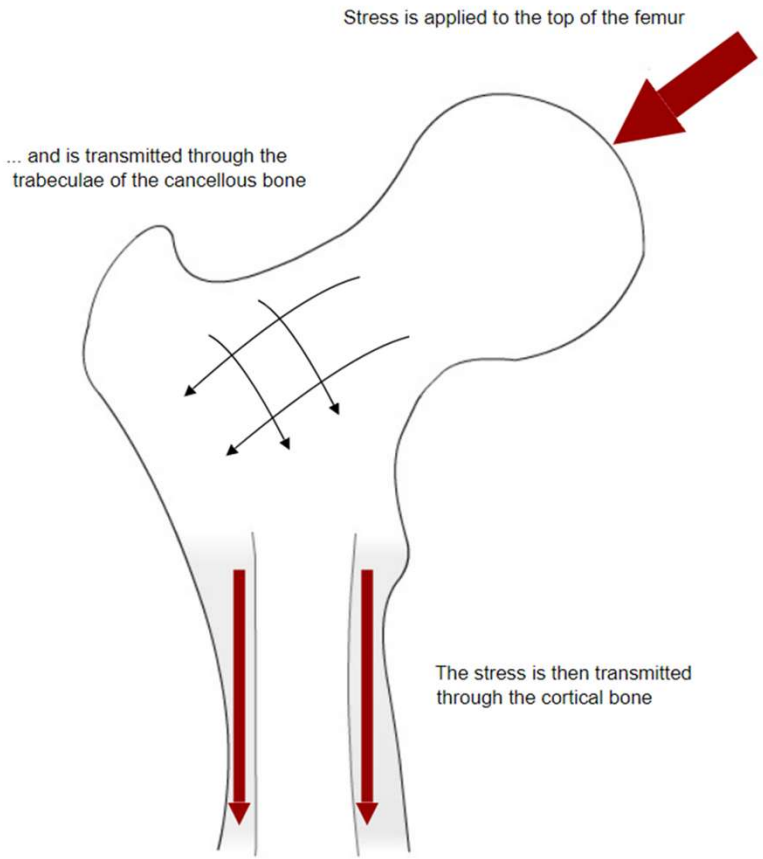
Bone tissue



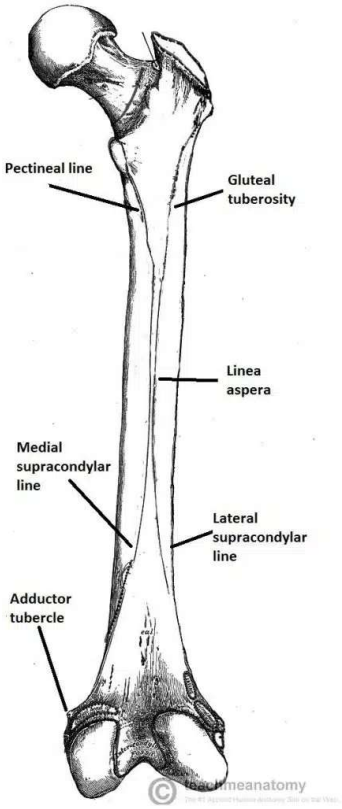


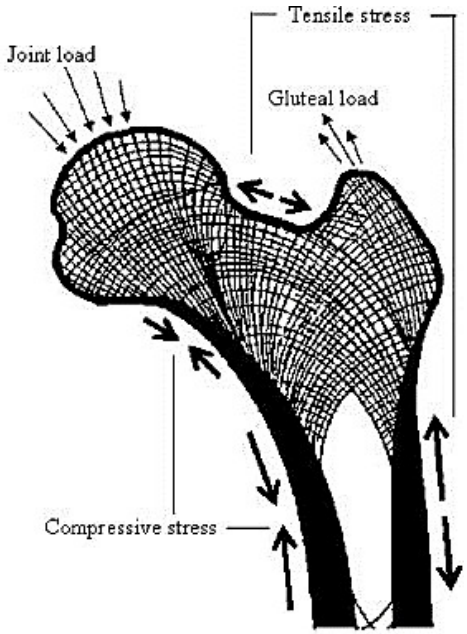
Cortical bone is highly dense and organized into a compact structure, with minimal spaces compared to spongy or cancellous bone (found inside bones).

Femur


Stress is applied to the top of the femur
... and is transmitted through the trabeculae of the cancellous bone
The stress is then transmitted through the cortical bone

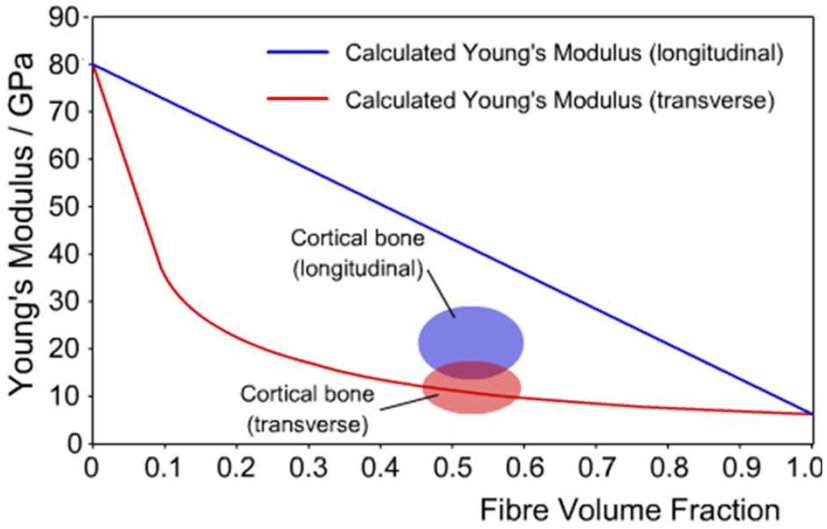
Femur and distribution of stresses





Mechanical properties of bone tissue

One can be considered a composite material consisting of collagen fibers and an inorganic matrix. The properties of a composite material (such as mechanical strength, hardness, density, etc.) can be calculated as a weighted average of the properties of its components. In the case of bone, the organic collagen provides flexibility and toughness, while the inorganic hydroxyapatite matrix contributes to stiffness and compressive strength.



In the longitudinal direction the difference is large, indicating the model does not give an accurate picture of the behaviour of bone. This difference occurs because the composite model of the microstructure of bone is highly simplified, since the collagen fibres are not aligned parallel to the axis of the osteons, and the bone mineral exists as discrete crystals, rather than forming a continuous matrix.

Mechanical properties of bone tissue

This table presents the **Young's Modulus (E)** for various materials relevant to bone, including collagen, bone mineral (hydroxyapatite), and cortical bone in two different directions (longitudinal and transverse).

Material	Young's Modulus, E (GPa)
Collagen (dry)	6
Bone mineral (Hydroxyapatite)	80
Cortical bone, longitudinal	11-21
Cortical bone, transverse	5-13

This indicates that collagen is quite flexible and not very stiff, which is expected given its role in providing toughness and flexibility to bone.

Hydroxyapatite is responsible for the bone's ability to resist compressive forces.

Cortical bone (compact bone) is stiffer when measured along the longitudinal direction (parallel to the bone's length). This direction aligns with the bone's natural load-bearing axis.

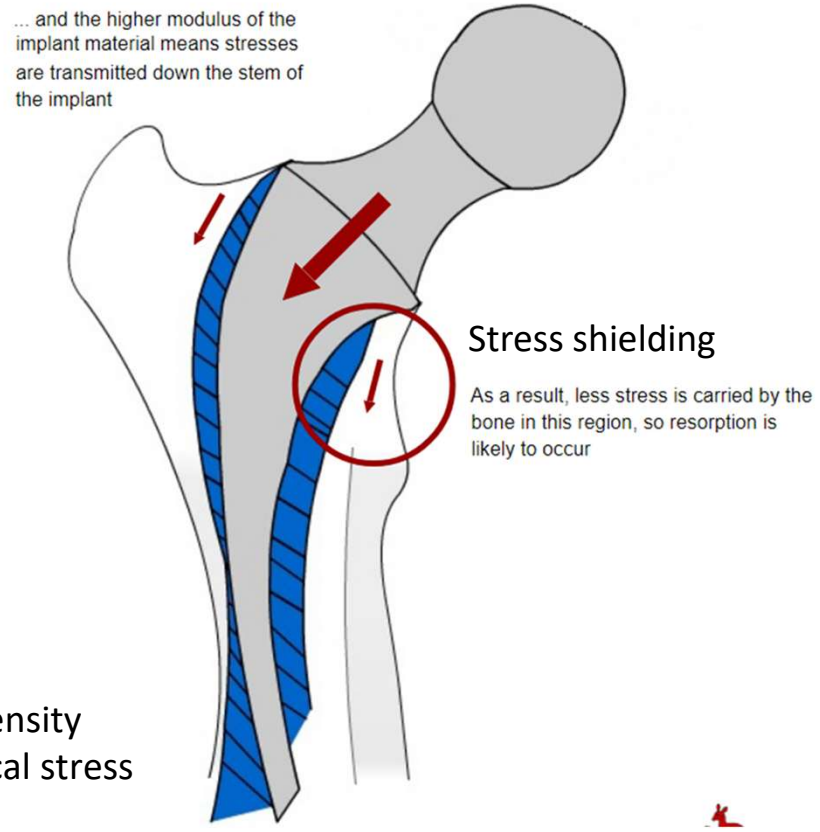
This reflects the anisotropic nature of bone

Mechanical properties of bone tissue

Different bones in the body are designed to withstand different forces depending on their location and function. For example, long bones like the femur and tibia, which bear most of the body's weight, are structured to resist high compressive forces, so they have high compressive strength.

	Longitudinal direction	Transverse direction
Tensile strength (MPa)	60-70	~50
Compressive strength (MPa)	70-280	~50

Stress shielding

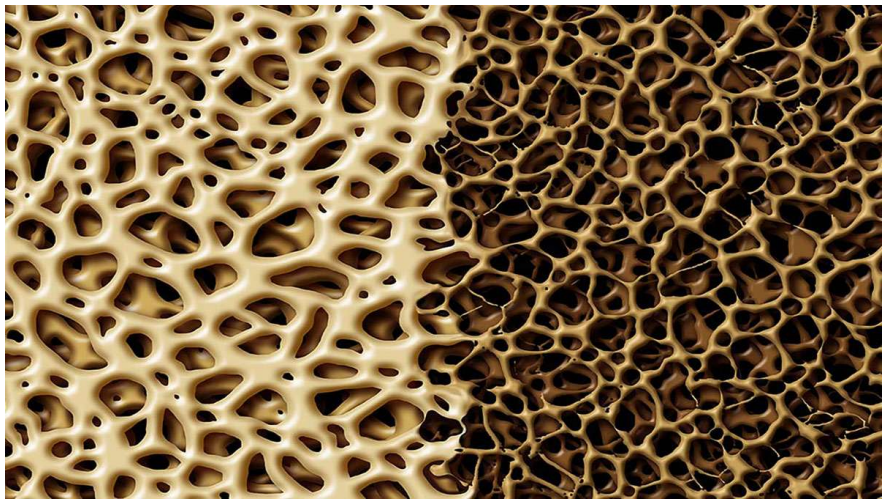


Stress shielding is the reduction in bone density (osteopenia) as a result of removal of typical stress from the bone by an implant

Bone remodelling

There is a continuous process in which bones are constantly broken down and rebuilt through the activity of two types of specialized cells: **osteoblasts** and **osteoclasts**.

The combined activity of osteoblasts (building) and osteoclasts (breaking down) allows bones to adapt to mechanical stresses, **repair microdamage**, and maintain healthy bone structure throughout a person's life.



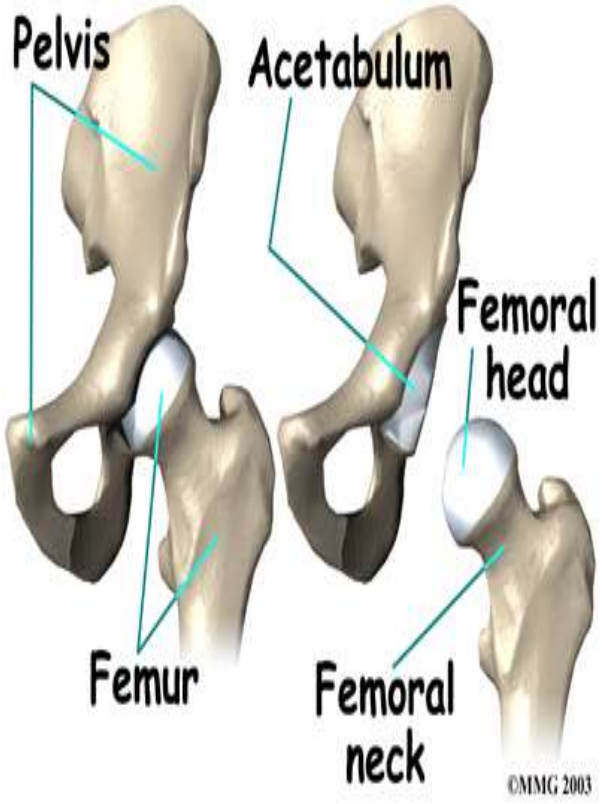


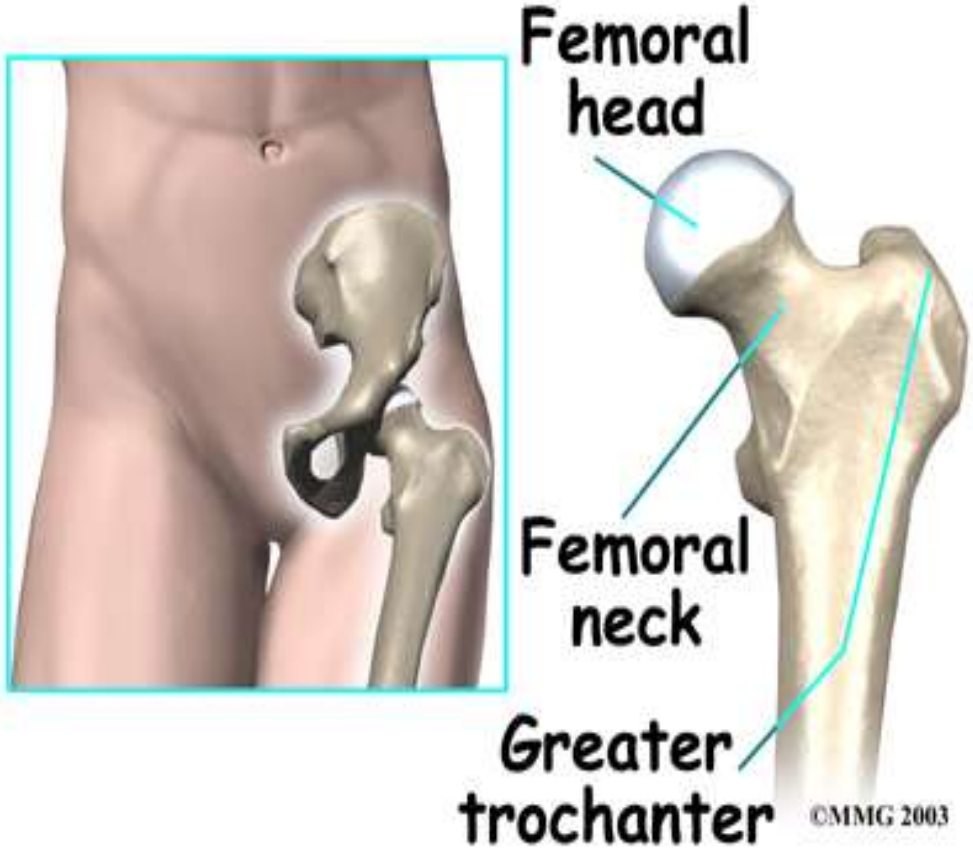
Osteoclasts are cells that break down bone tissue.

Osteoblasts are cells responsible for forming new bone tissue.

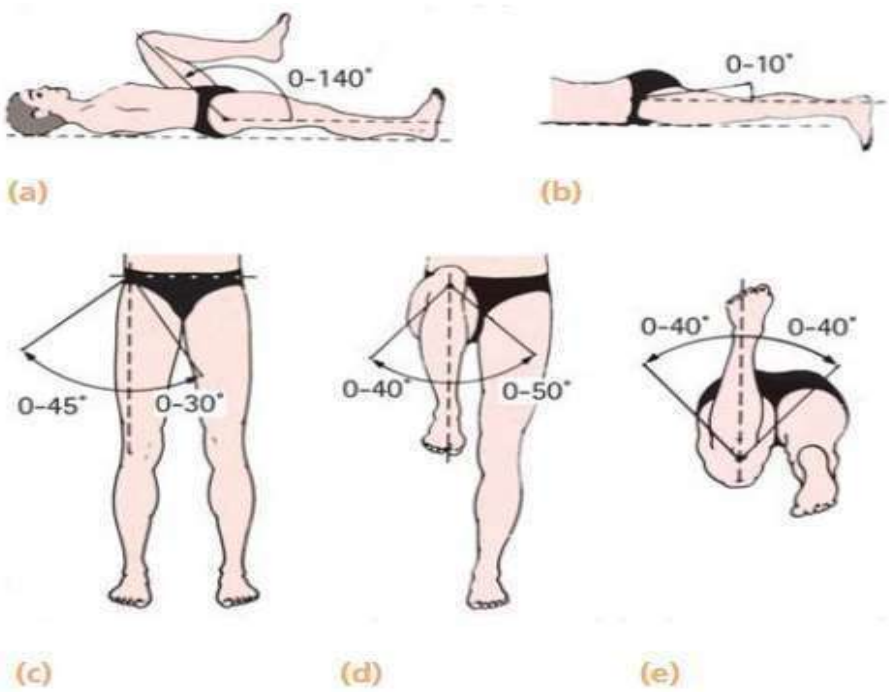
Hip anatomy and fundamental biomechanics

Anatomy—Hip


Pelvis
Acetabulum
Femoral head
Femur
Femoral neck
©MMG 2003


Femoral head
Femoral neck
Greater trochanter
©MMG 2003

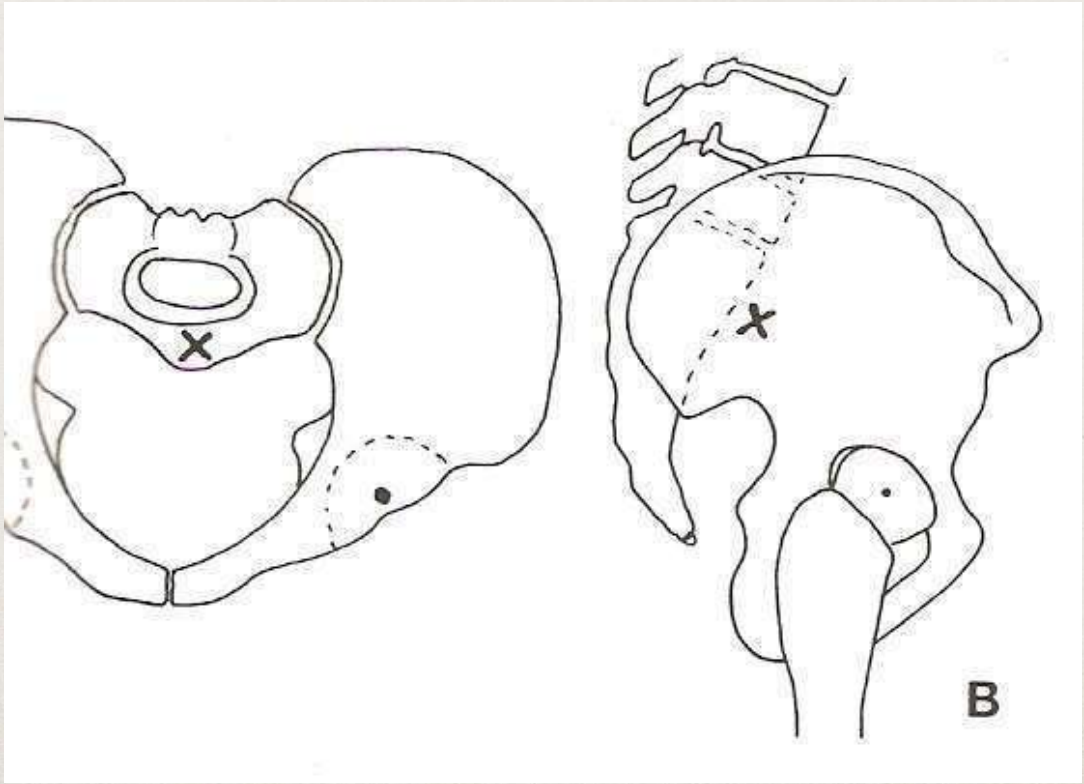
Normal ROM at hip..



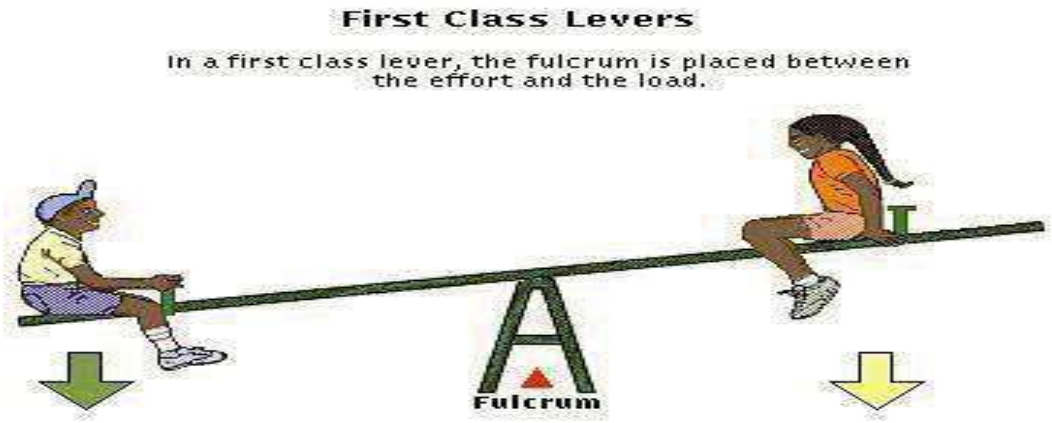
19.6 Normal range of movements (a) The hip should flex until the thigh meets the abdomen, but (b) extends only a few degrees. (c) Abduction is usually greater than adduction. The relative amounts of internal and external rotation may vary according to whether the hip is in (d) flexion or (e) extension.

Centre of gravity

In humans- just anterior to S2



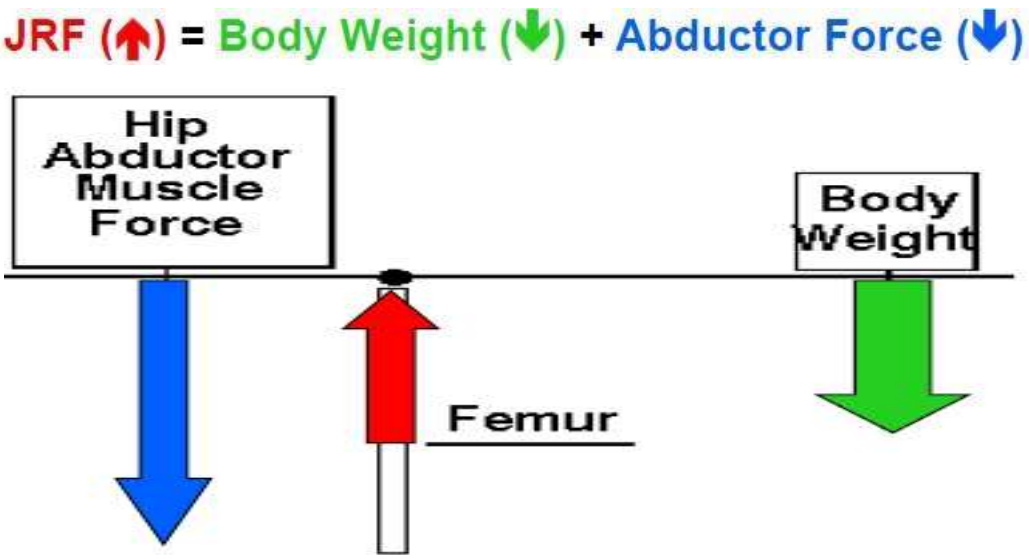
- **First order lever**

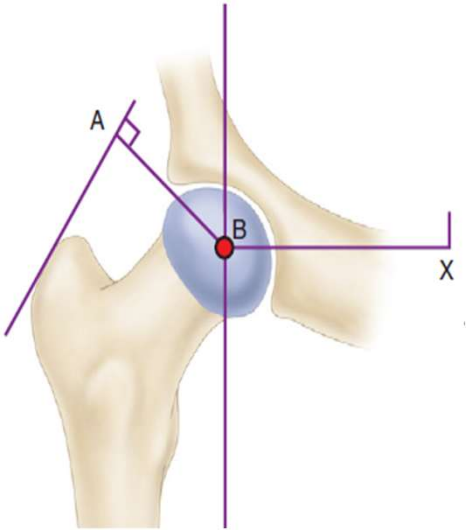


fulcrum (hip joint)
forces on either side of fulcrum
i.e, body weight & abductor tension

Joint reaction force

- Defined as force generated within a joint in response to forces acting on the joint
- In the hip, it is the result of the need to balance the moment arms of the body weight and abductor tension
- Maintains a level pelvis

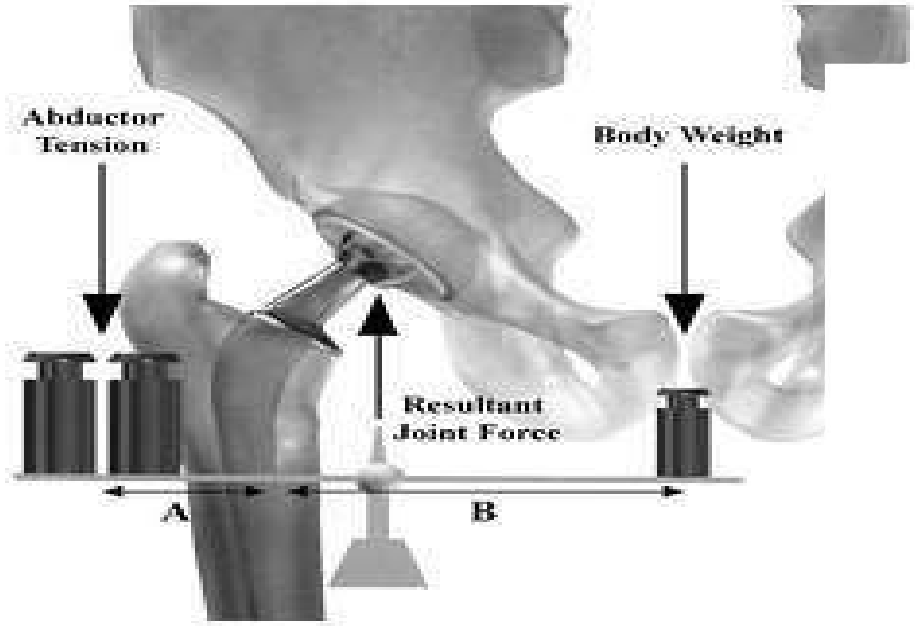


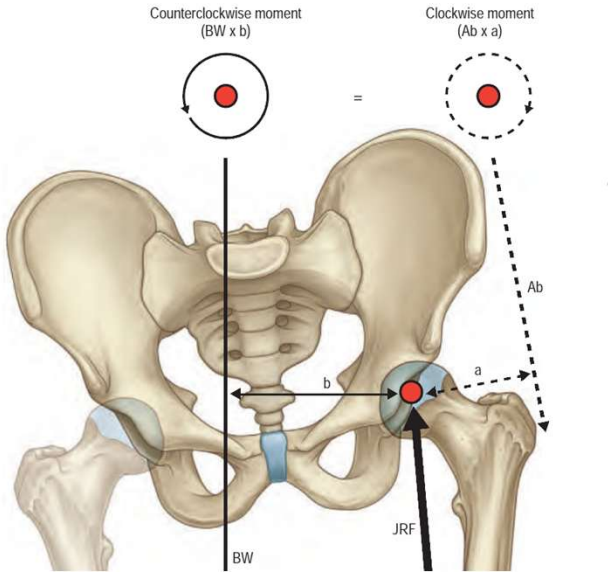


Biomechanics

- To maintain stable hip, torques produced by the body weight is countered by abductor muscles pull.

Abductor force X lever arm1 = weight X leverarm2





- The force of the abductor muscles also creates a moment around the centre of the femoral head;
- however this moment arm is considerably shorter than the effective lever arm of body weight. Therefore the combined force of the abductors must be a multiple of body weight
- The magnitude of the forces depends critically on the lever arm ratio, which is that ratio between the body weight moment arm and the abductor muscle moment arm (a:b).
- Typical levels for single leg stance are three times bodyweight, corresponding to a level ratio of 2.5.

Anything that increases the lever arm ratio also increases the abductor muscle force required for gait and consequently the force on the head of the femur.

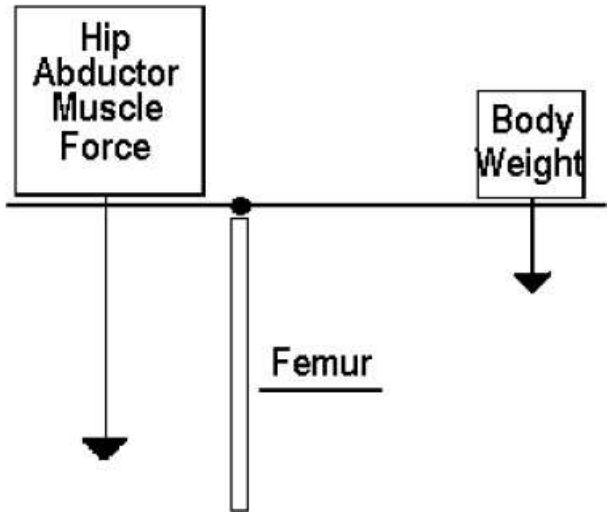
- Short femoral necks
- Wide pelvis

This tendency means that women have larger hip forces than men.

Normal hip

- Determinants of JRF (Key elements in hip biomechanics):

•Body weight
•Body weight moment arm
•Abductor force (muscles)
•Abductor force moment arm

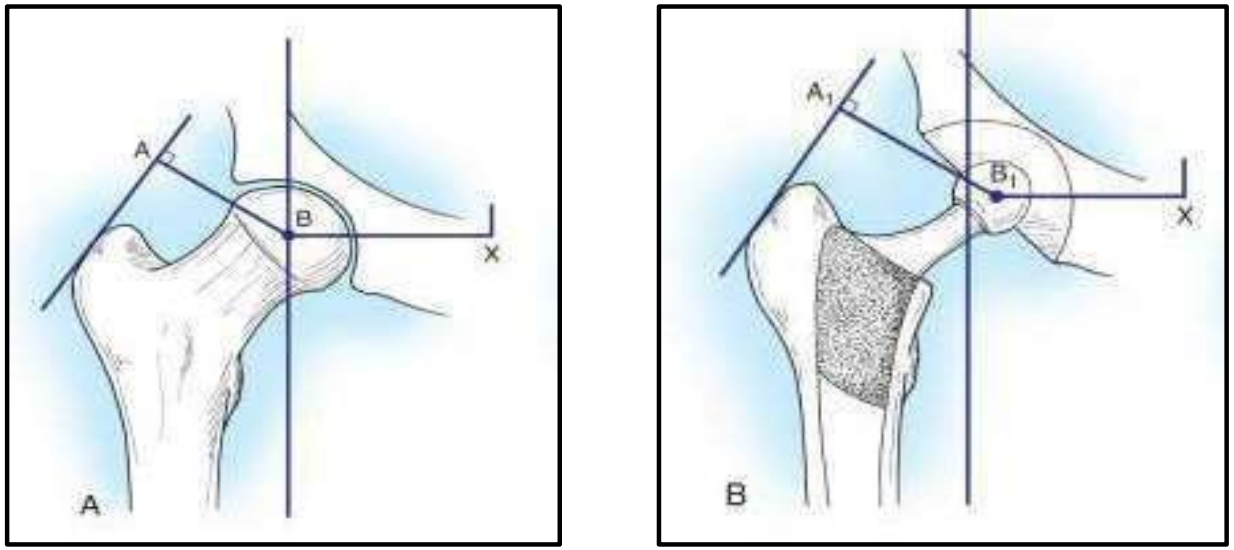


- **Forces generated within a joint in response to external forces (both intrinsic and extrinsic).**
  - Can reach 3 to 6 times body weight & is primarily due to contraction of muscles crossing the hip.
  - Twice during SLRT
  - 3 times in single leg stance
  - 5 times in walking
  - Upto 10 times while running

  - Reduced to half upon using a cane

Applied biomechanics in THA

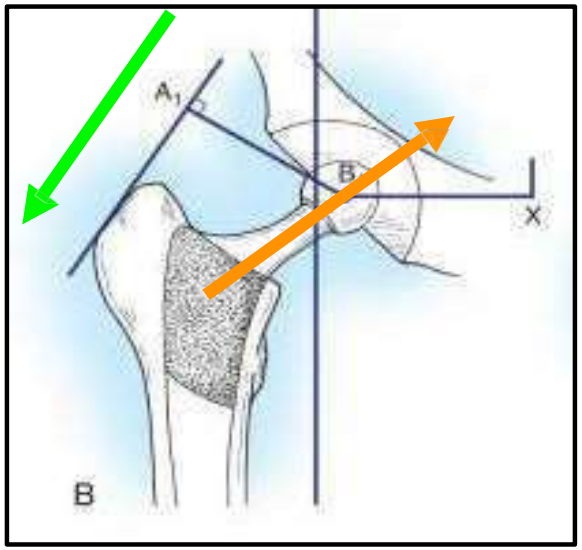
Principle- To decrease joint reaction force



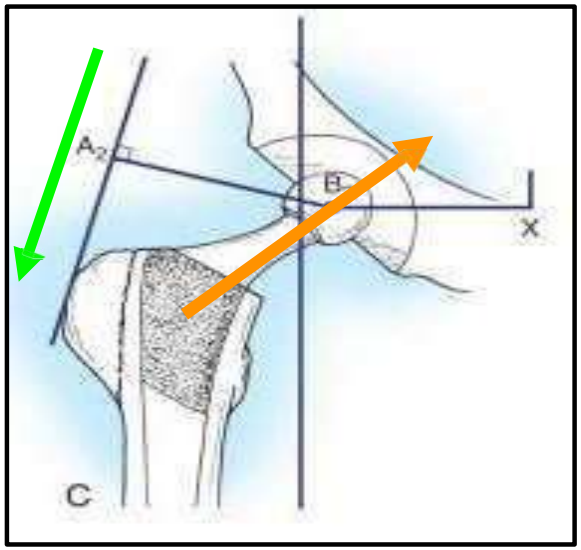
Centralisation of the femoral head by deepening of acetabulum

- decreases BW lever arm

Decreased BW lever arm



Lengthened abductor lever arm



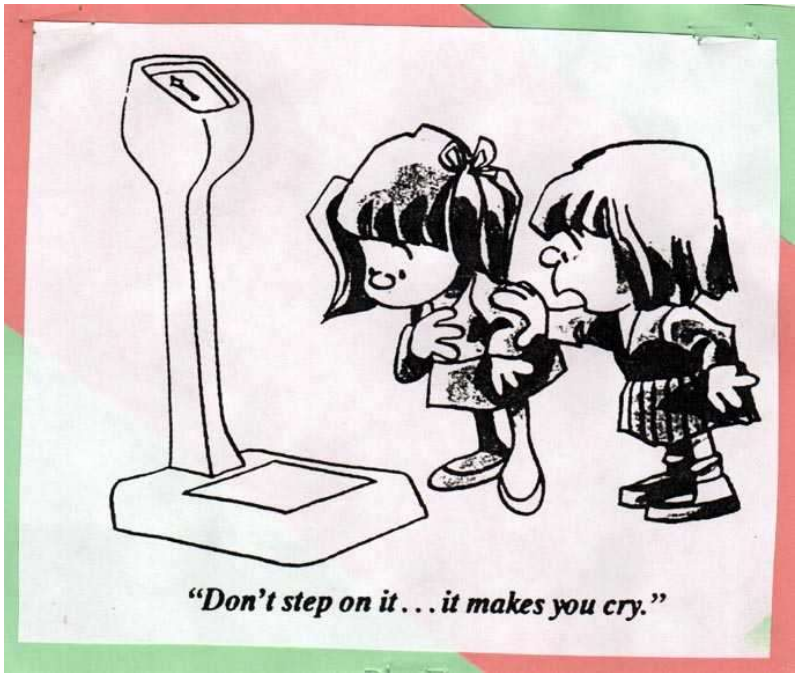
Reduced wear of implants

Weight Gain

Body weight and lever arm increases.

Abductor muscular forces are to be increased to counteract body weight.

Increased joint forces across the joint leading to increased degeneration



Indications

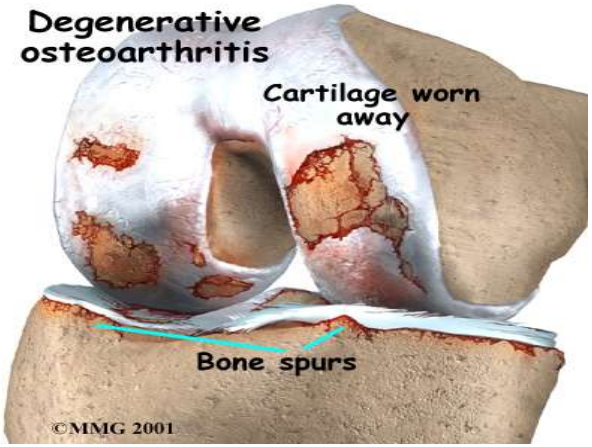
Arthritis(degenerative joint disease, rheumatoid arthritis)

Femoral neck fractures

Failure of previous reconstructive surgeries(failed prosthesis, osteotomy)

Problems resulting from congenital hip disease





Osteoarthritis of Hip

Contact area is decreased, secondary to deformity & loss of sphericity of femoral head.

This concentrates compressive forces across the hip joint to smaller area resulting in increased pressure or load.

In hip disease patient decreases this load by *tilting towards the affected side in stance phase* → decreasing partial body weight lever arm

Osteoarthritis of Hip

- For most painful hip joints, however, the reductions in compression generally required are greater than *can be realistically achieved through weight loss*.

- The solution must be in a reduction of *abductor muscle force* requirements.

Principles of Osteotomy

Osteotomies improve hip function

- Increasing contact area / congruency
- Improve coverage of head
- Moving normal articular cartilage into weight bearing zone
- Restore biomechanical advantage / Decreasing joint reactive forces
- Stimulating cartilage repair

Total hip replacement

HISTORY OF HIP REPLACEMENT SURGERY

Total Hip replacement is the replacement of a severely damaged hip with an artificial joint.

Total hip arthroplasty (THA) has completely revolutionized the nature in which the arthritic hip is treated, and is considered to be one of the most successful orthopaedic interventions of its generation.

In 1891, Professor Themistocles Glück presented the use of ivory to replace femoral heads of patients whose hip joints had been destroyed by tuberculosis.



HISTORY OF HIP REPLACEMENT SURGERY

A British Orthopedician, Pioneer of modern hip replacement Arthroplasty.

Developed the techniques of THR in 1960s.

It consisted of three parts;

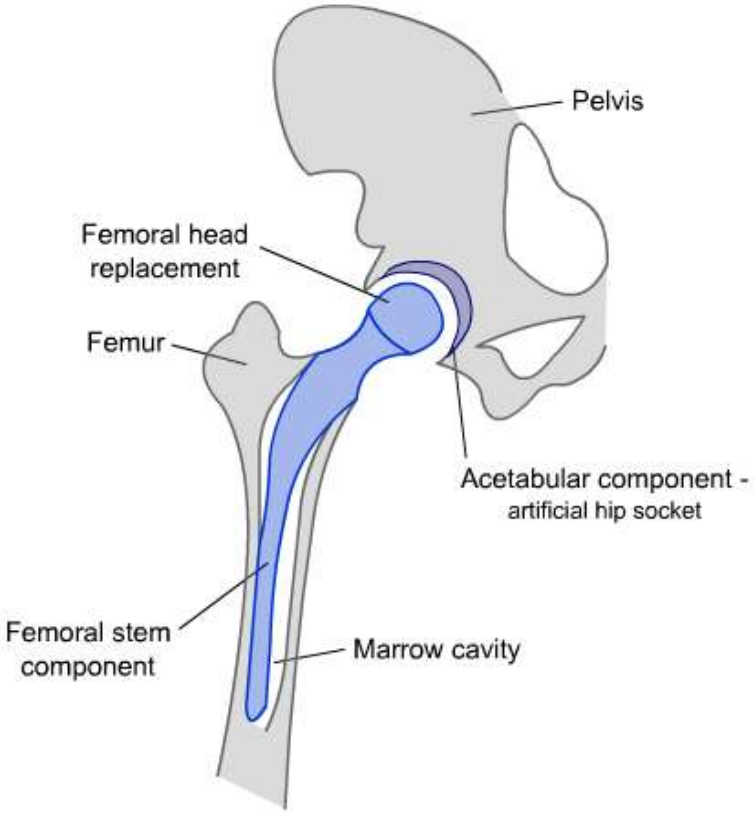
a metal femoral stem, a polyethylene acetabular component and acrylic bone cement - which was borrowed from dentists.

It was called the *low friction arthroplasty* as Charnley advocated the use of a small femoral head which reduces wear due to its smaller surface area.



Professor John Charnley
(1911-1982)

THR





Femur with cemented titanium alloy prosthesis.
(S. Blatcher, PhD Thesis, Queen Mary College, London, 1995)

THA Implants



The hip prothesis

The **femoral stem** is inserted into the shaft of the femur, properly positioning the body of the implant. The femoral stem essentially replaces the bone within the femur and has a structural role that needs to be as close as possible to the strength and durability of natural bones.



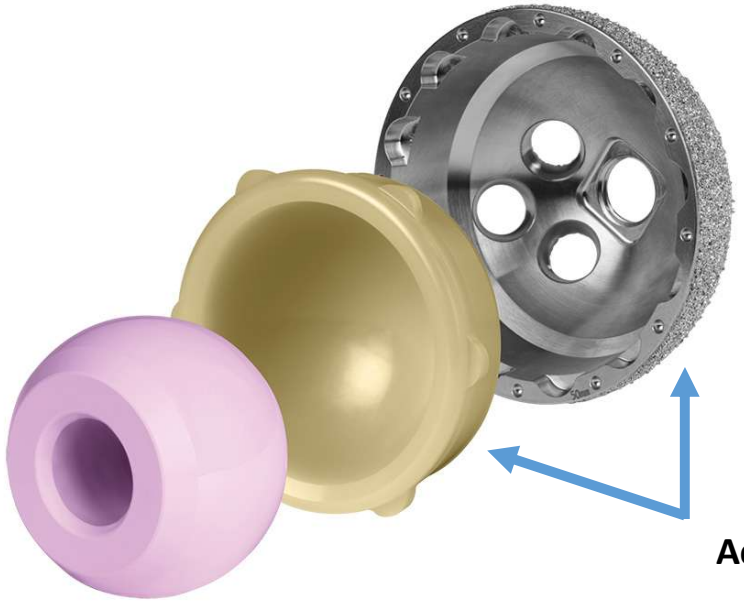
The hip prothesis

The **femoral head component** must meet certain requirements: it must have a specific shape that fits perfectly into the acetabular cup/hip socket, and its surface must have a low coefficient of friction to allow smooth movement.



The hip prothesis

The **acetabular cup** fits into the hip socket and takes over the role of the worn-out cartilage. As such, this component must have a low wear rate and, at the same time, minimize the wear of the femoral head component.



Acetabular cup

Four important variables determine the stability of THR

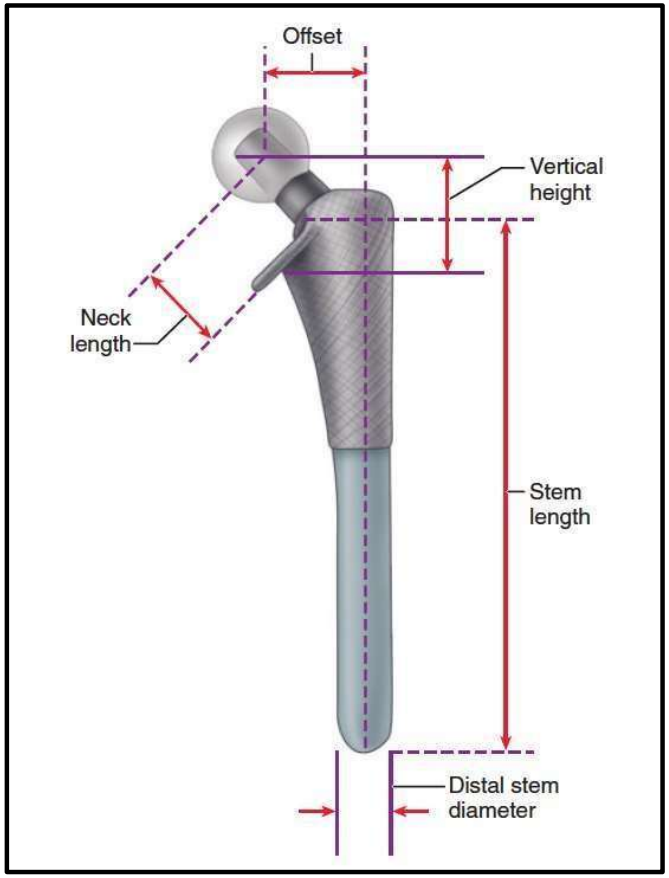
1. **Component design:** femoral head diameter, head–neck ratio, neck length, acetabular cup geometry, femoral stem geometry.
2. **Component position:** For the acetabular cup, important factors include: inclination, anteversion, medial/lateral position. For the femoral component: stem version, neck orientation, depth of insertion.
Poor positioning may cause: impingement, reduced range of motion, edge loading, increased wea dislocation.
3. **Soft tissue positioning** (restoration of offset): correct abductor muscle tension, appropriate muscle lever arm, joint stability, normal gait,
proper joint reaction forces
4. **Soft tissue function:** Important tissues include: abductor muscles, joint capsule, tendons, surrounding stabilizing muscles.
The abductors are especially important

Offset

Medial or horizontal offset
Centre of the head to the axis of the stem.

Vertical offset(height)
Determined by the base length of the prosthetic neck and length gained by head.

The depth the implant is inserted into the femoral canal alters the vertical height.



HEAD Diameter

AFFECTS ROM ,IMPINGEMENT,STABILITY OF ARTICULATION.

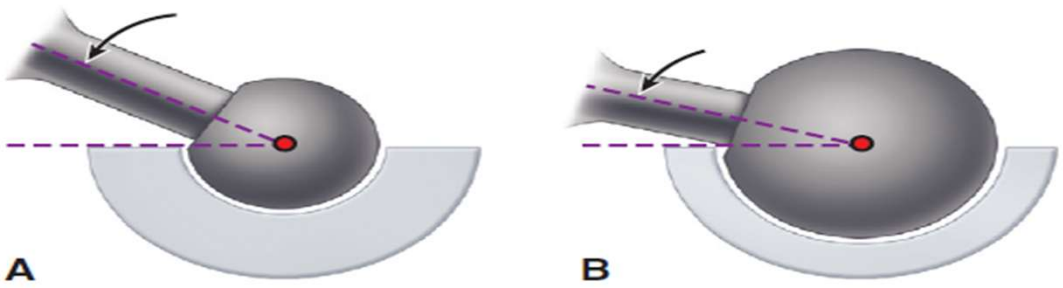


FIGURE 3-11 Range of motion with different head sizes. For given diameter neck, implant with smaller femoral head (**A**) will have lesser arc of motion than larger one (**B**).

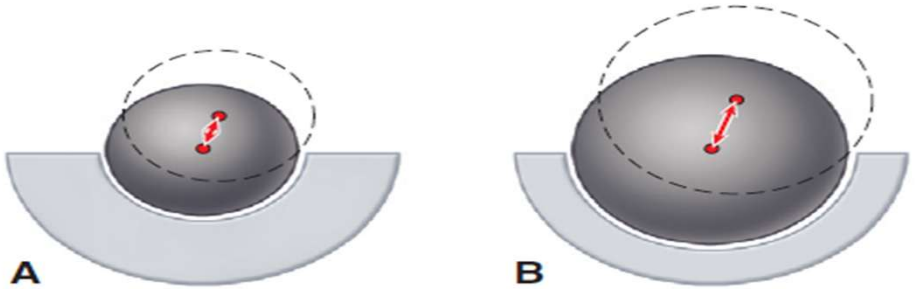


FIGURE 3-12 Jump distance. With subluxation, smaller head (**A**) has shorter distance to travel before escaping rim of acetabular component than larger one (**B**).

Range of Motion

Heavily influenced by prosthesis design



Fig. 2.2 Contemporary and historical femoral components documenting the wide variety of designs with head size diameters ranging from 62 to 22 mm (*left to right*: Zweymüller, Exeter, Corail, St. Georg, Silent & Resurfacing, CFP, Meta, Charnley)



Fig. 2.3 View of the backside of new and revised acetabular components; from left to right (*columns*): press-fit uncemented cups, cemented cups, threaded (*bottom*: expansion) cups, monobloc cups

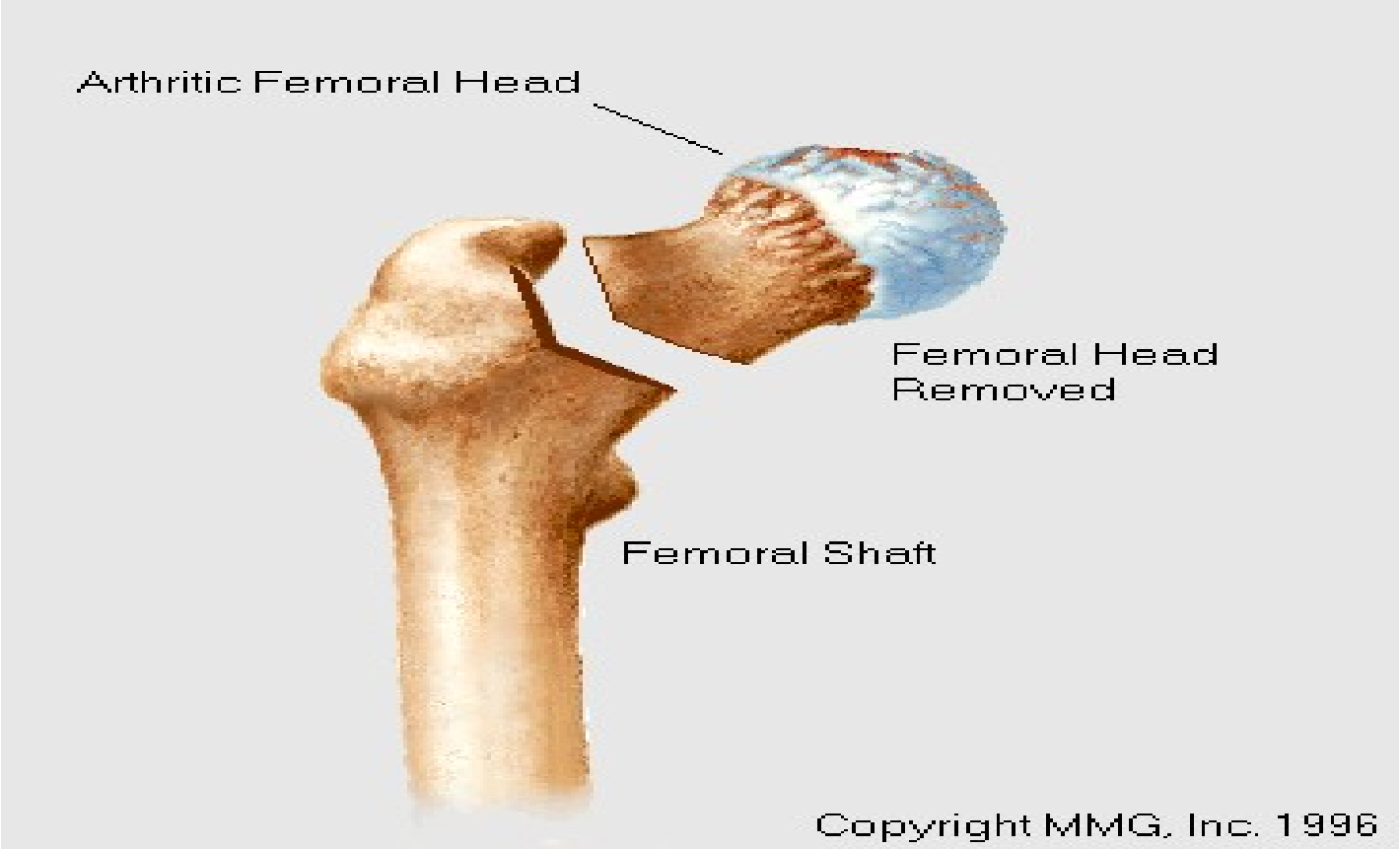
So how large it has to be…

Head size	Dislocation rates
22mm	Upto 18 %
28mm	0.6-3%
32mm	0.5%
38mm	0.0%

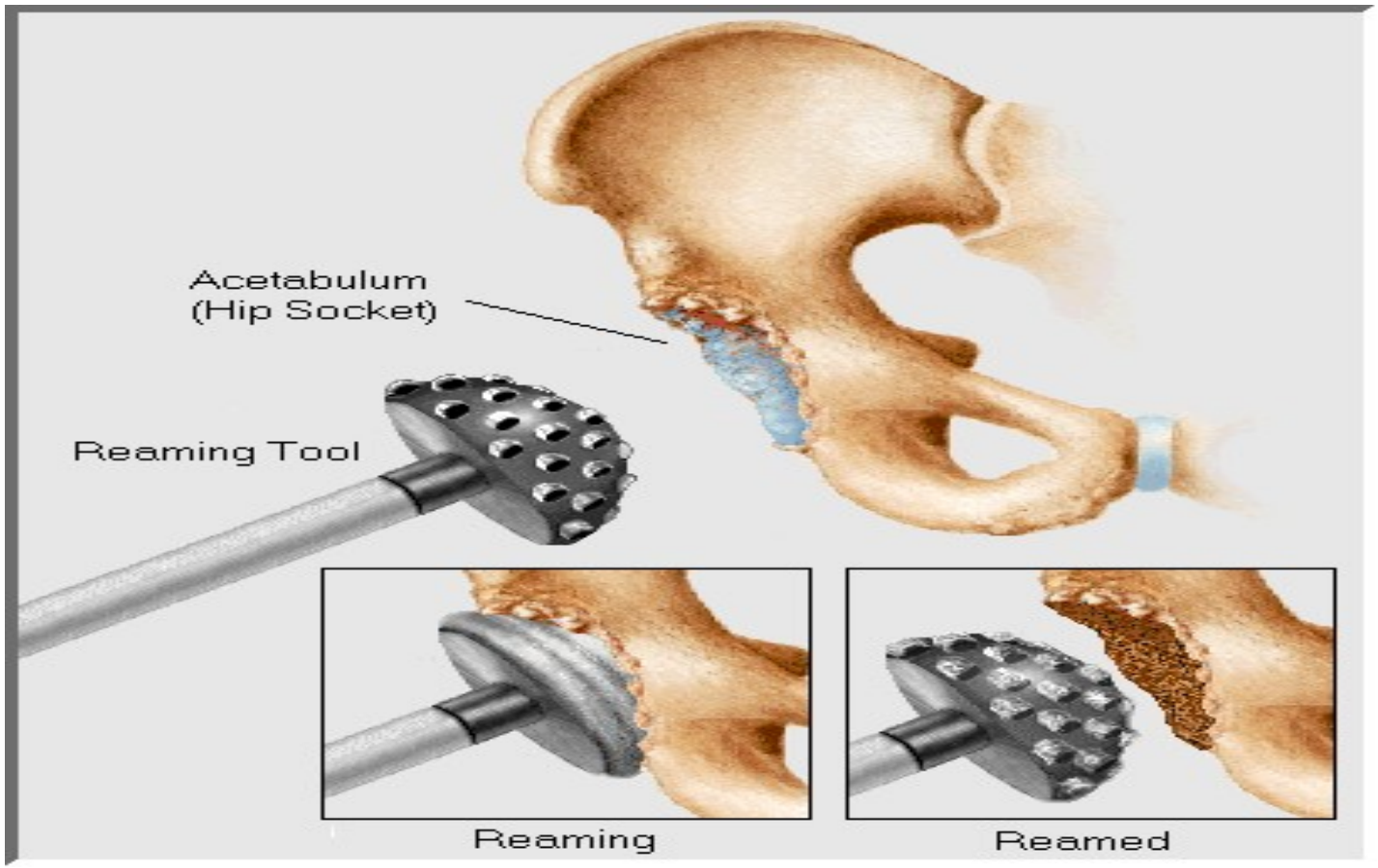
Size is not the only parameter for dislocation tho…
Implant position and soft tissue tension is equally important

Surgical sequence

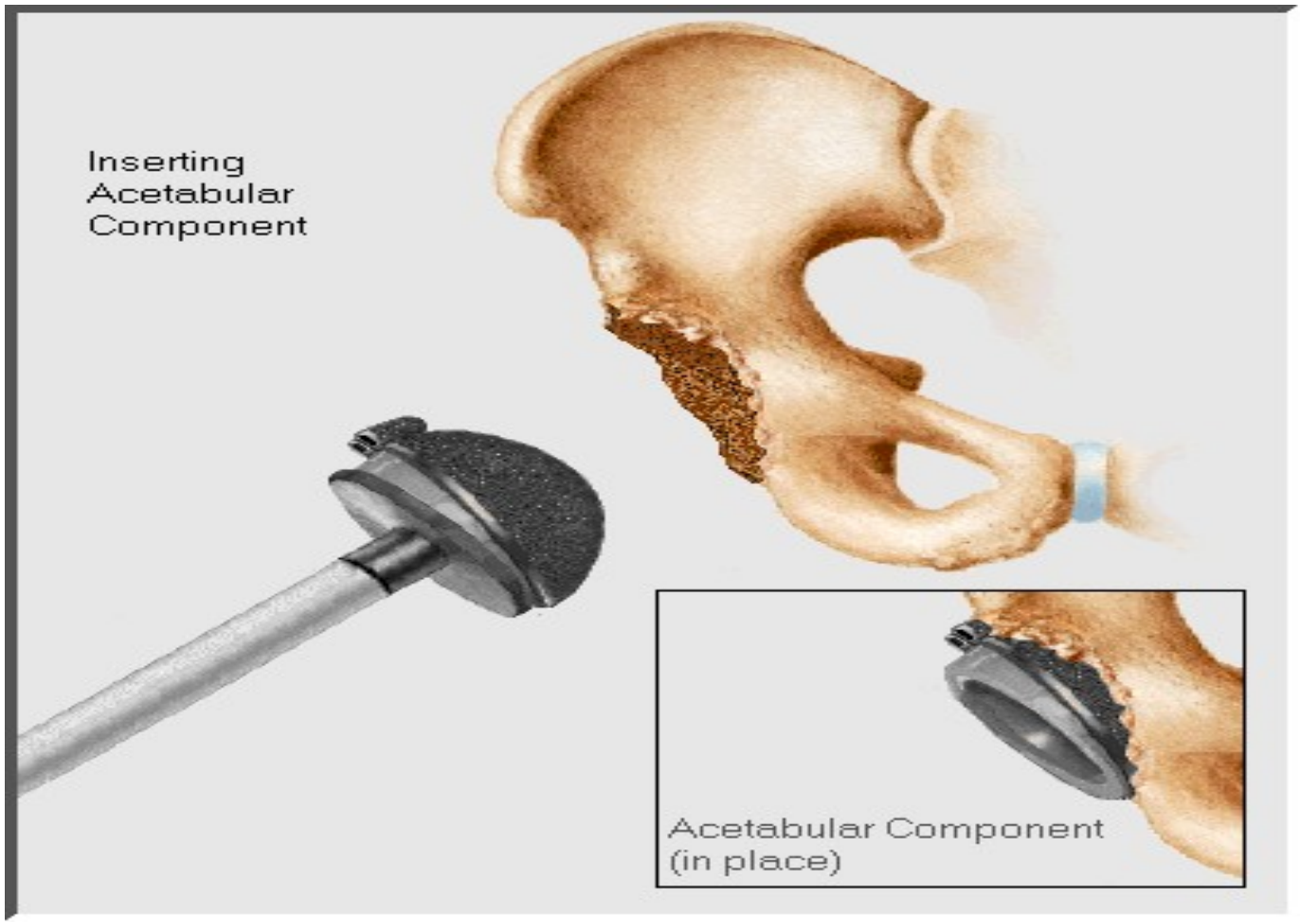
Femoral neck resection



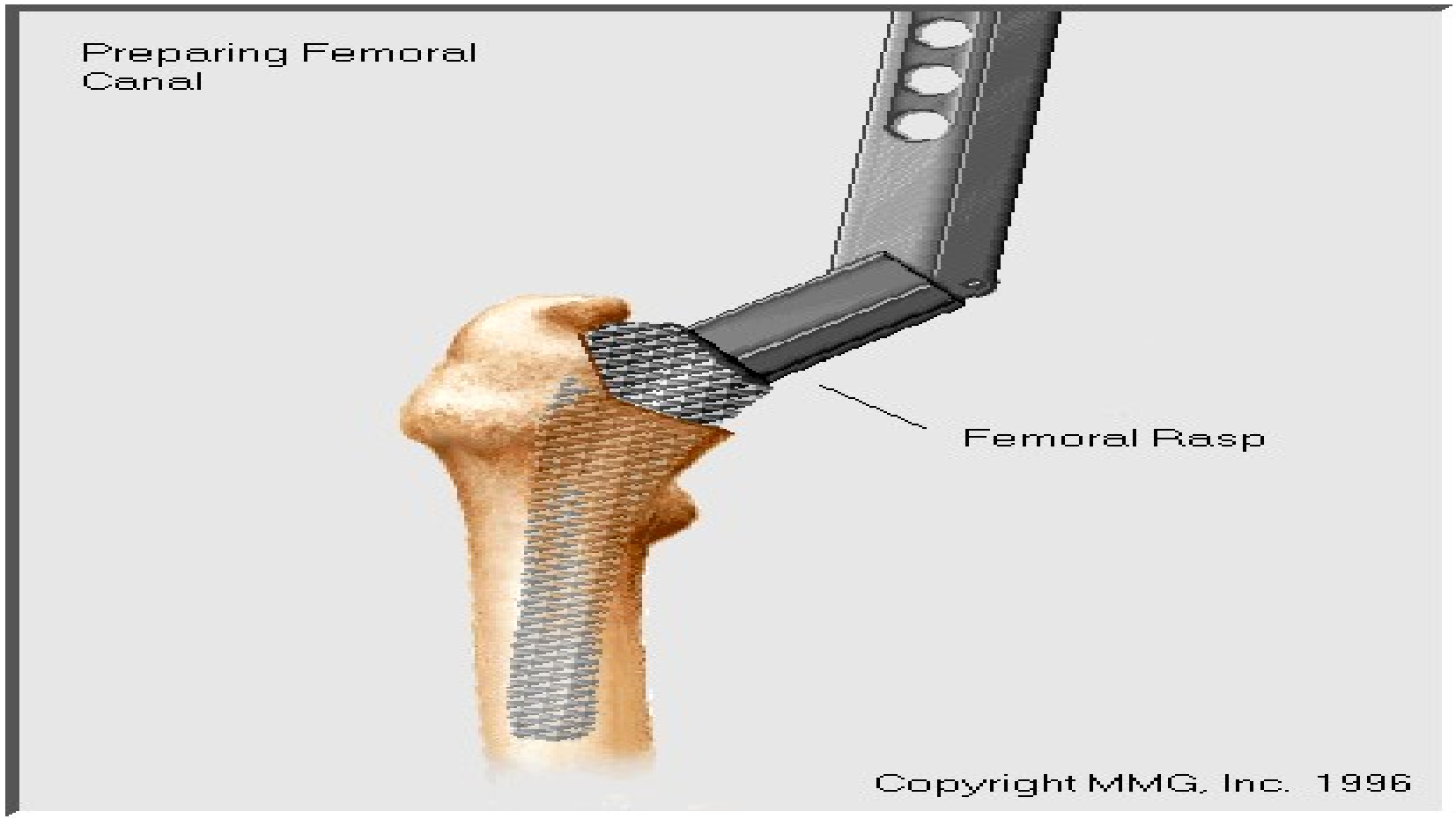
Acetabular reaming



Insertion of acetabular component



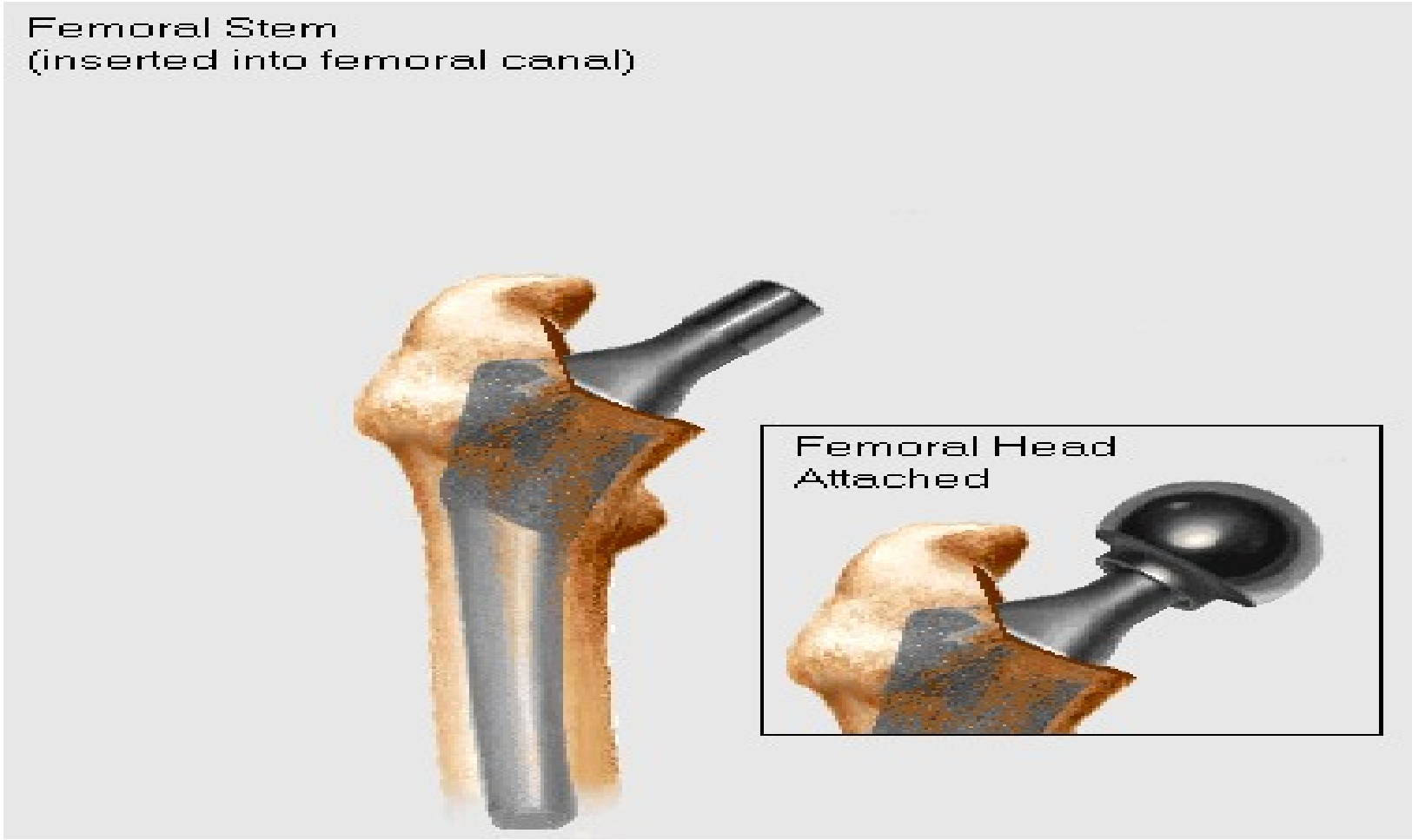
Reaming/broaching of femoral component



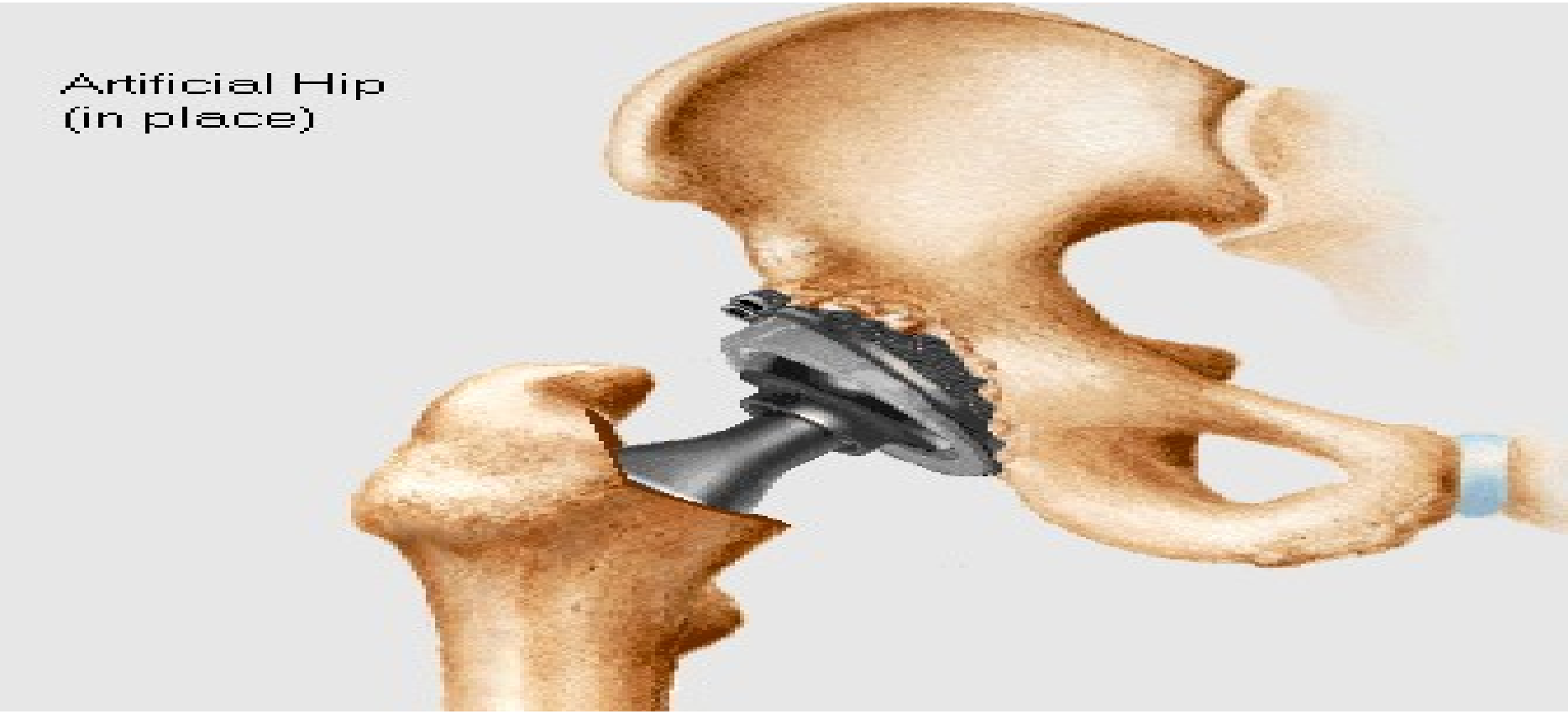
Insertion of femoral component



Femoral head impaction



Final implant



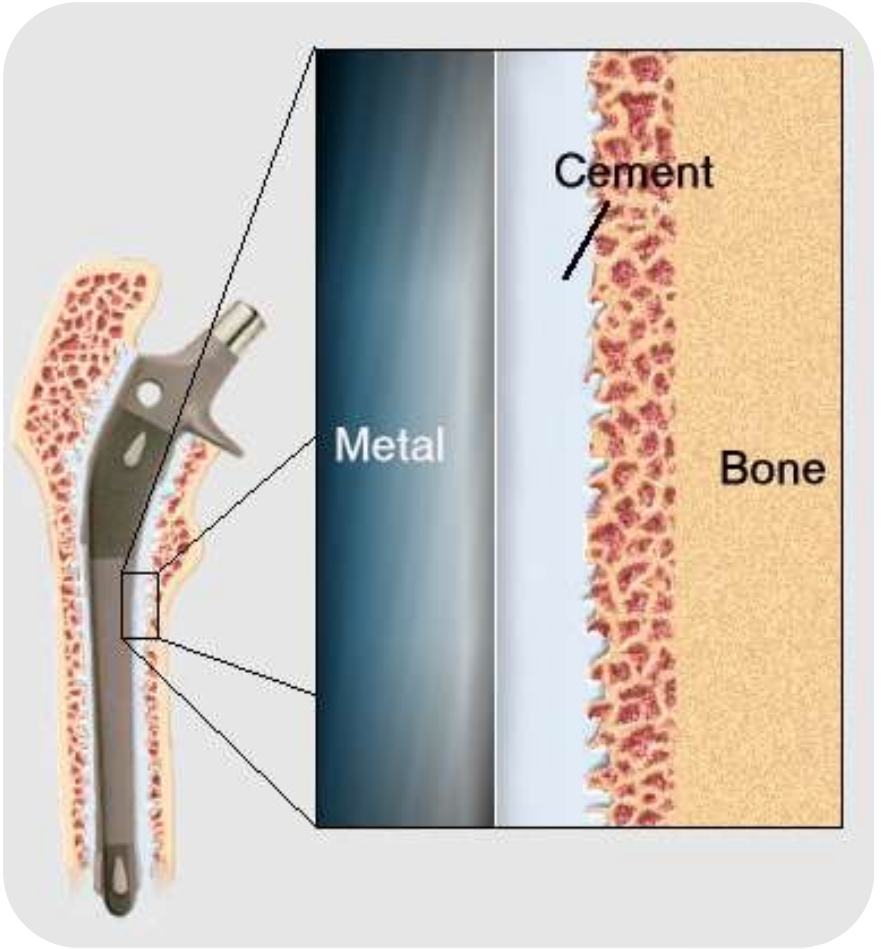
Total Hip Replacement



Fixation concepts

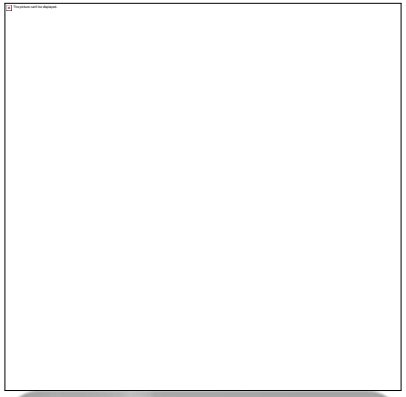
Implant Choice

Cemented joint replacement (cemented joint arthroplasty) - A procedure in which bone cement or polymethylmethacrylate (PMMA) is used to fix the prosthesis in place in the joint



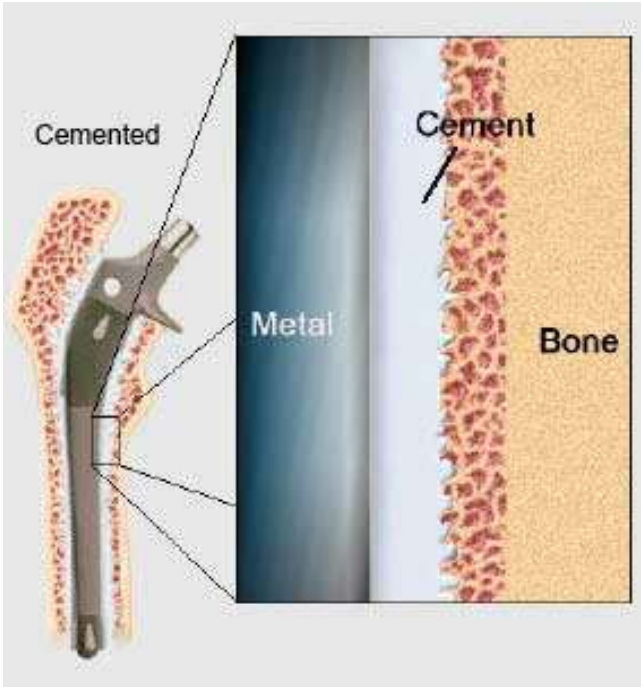
Cemented:

- Elderly (>65)
- Low demand
- Better early fixation
- late loosening



Cement fixation

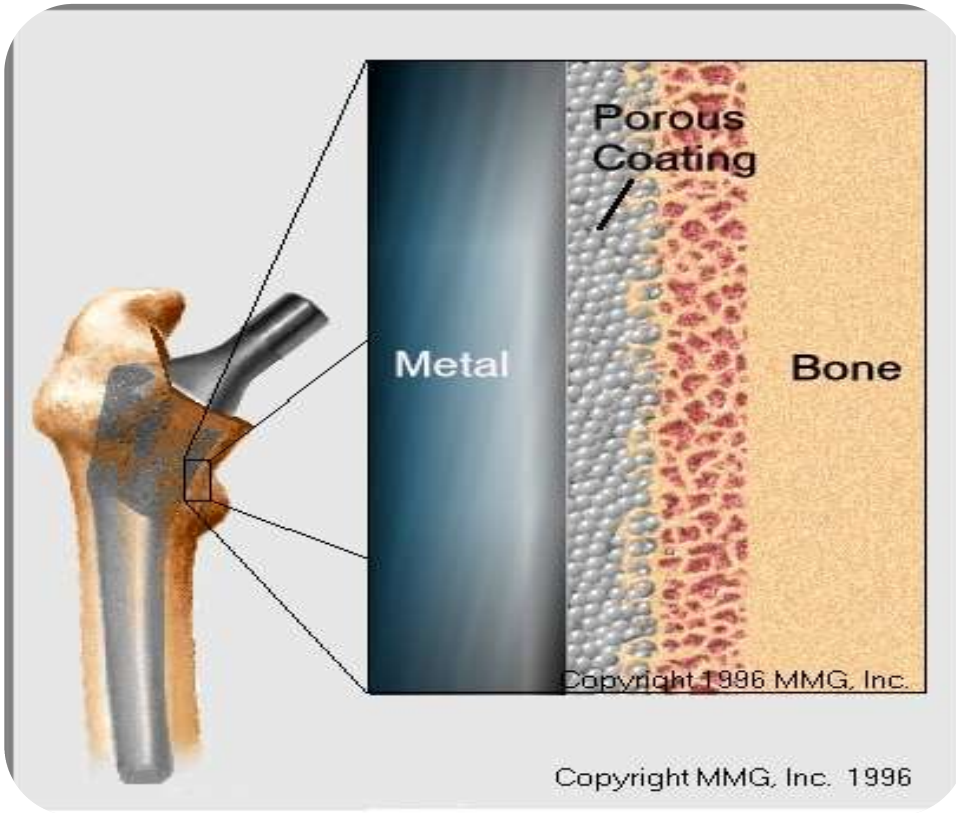
Polymethylmethacrylate (PMMA) is widely used as bone cement to fix the femoral stem to the bone in total hip replacements. PMMA works by filling the space between the implant and the bone, effectively securing the implant in place. During the curing process (polymerization of PMMA), the material releases heat (**exothermic reaction**). This **heat can cause cell death** (thermal necrosis) in the surrounding bone tissue, which is a recognized drawback of using PMMA bone cement.



Cementless

A procedure that does not involve bone cement to fix the prosthesis in place; an anatomic or press fit with bone ingrowth into the surface of the prosthesis leads to a stable fixation; this procedure is based on a fracture-healing model.

Hydroxyapatite (HAP) Coating: **Hydroxyapatite (HAP)** is a ceramic material whose **chemical composition closely resembles that of natural bone mineral**. Coating the implant with hydroxyapatite encourages **osteointegration** by promoting the growth of new bone tissue on the surface of the implant. The coating helps to accelerate and enhance the bond between the implant and the bone.



Cementless:

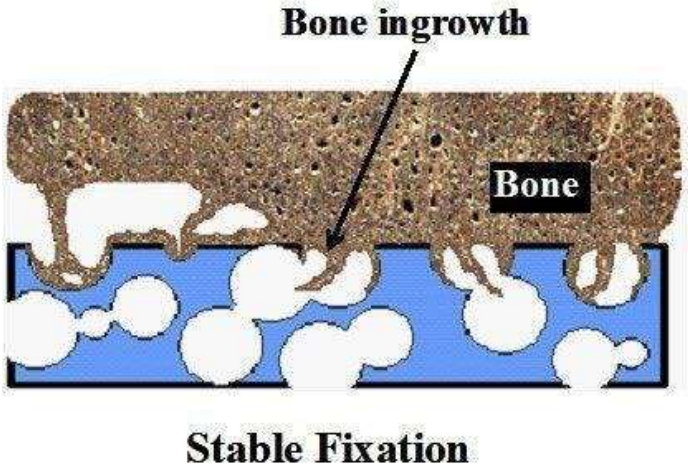
- Younger
- More active
- Protected weight-bearing first 6 weeks
- Better long-term fixation



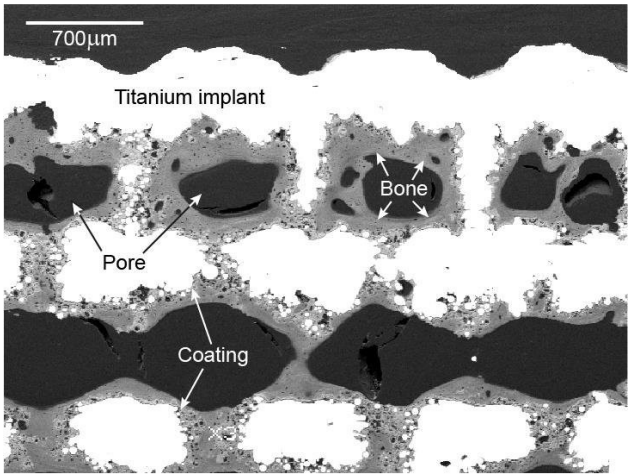
Alternative Fixation: Using porous materials

An alternative to using bone cement is to design implants with **porous surfaces**. These **porous structures** allow **bone ingrowth**, where the bone tissue grows into the pores of the implant, creating a strong **biological fixation**. This method is called **cementless fixation**





Stable Fixation



Femoral component designs & Materials

The hip prothesis

Biological Environment
The human body is not an easy environment for a material to function in for an extended period. The material must be able to operate for years at a temperature of 37°C in a highly humid and corrosive environment.

Tissue-Implant Interaction
Implant materials should be designed to minimize the adverse effects that a foreign substance introduced into the body might cause. The immune system typically attacks anything originating from outside the body, which can lead to inflammation. High levels of certain metals in the bloodstream can cause various issues, including cytotoxicity and cancer development. Therefore, it is crucial to choose materials that have minimal negative effects on the body.

Cost
As with most areas in materials science, cost is a significant factor in material selection. Manufacturers often need to find an appropriate balance between the material's performance and its cost.

Materials selection for hip replacement

Requirements from stem: mechnical properties, biocompatibility, longevity

Options:

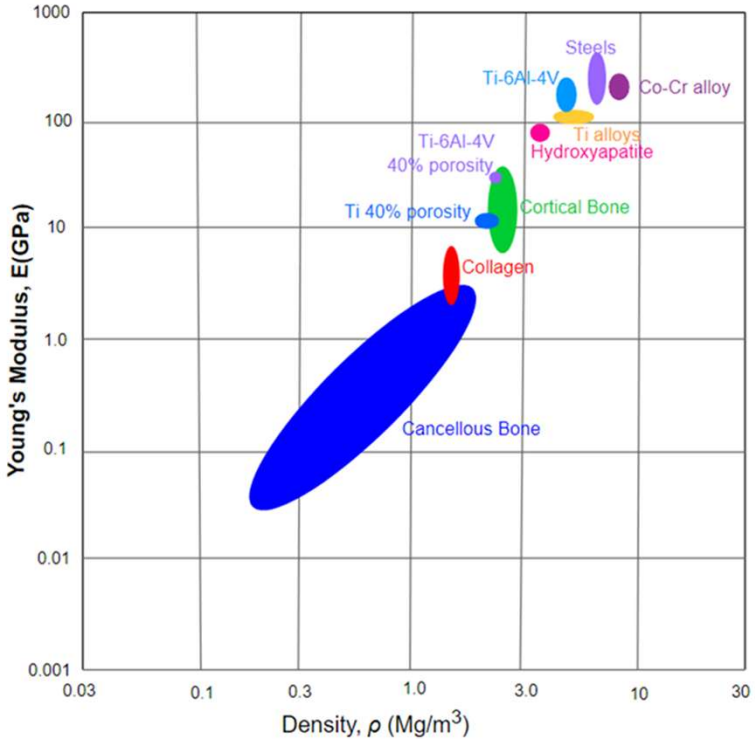
Polymers: low mechanical strength

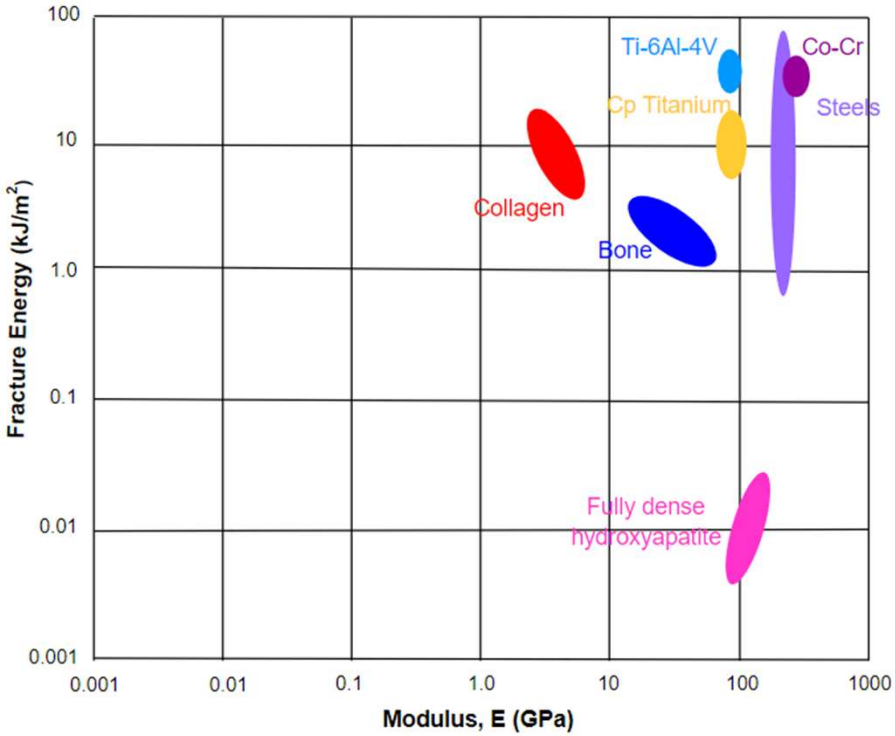
Metals: can have corrosion problems, but highmechanical strength

Ceramics: fragile

Elasticity modulus is critical for the material
that is to be selected for stem construction.

Materials selection for hip replacement





Materials selection for hip replacement

Stainless steel: Fe, Cr, Ni, Co alloy

High elasticity modulus of stainless steel accounts for stress shielding.

Nickel can be allergic

Cost: relatively low cost



Materials selection for hip replacement

Co-Cr alloys: Co-Cr-Mo

Cobalt-chromium-molybdenum (CoCrMo) alloy is one of the most used metals in total hip replacement (THR) due to the alloy's superior corrosion qualities and biocompatibility.

High elasticity modulus results in stress shielding

Production cost is higher than stainless steel



Materials selection for hip replacement

Titanium alloys

Titanium allys are **bioinert**. This property minimizes the risk of the body rejecting the implant

Titanium alloys have better **creep resistance** than traditional stainless steels

Titanium alloys are five Times More Expensive Per Kilogram Than Steel

Elasticity modulus (108 GPa) is half of steel: good for stress shielding but not enough

Titanium alloys exhibit **excellent corrosion resistance** due to the formation of a passive oxide layer (TiO_2) on their surface

Materials selection for hip replacement

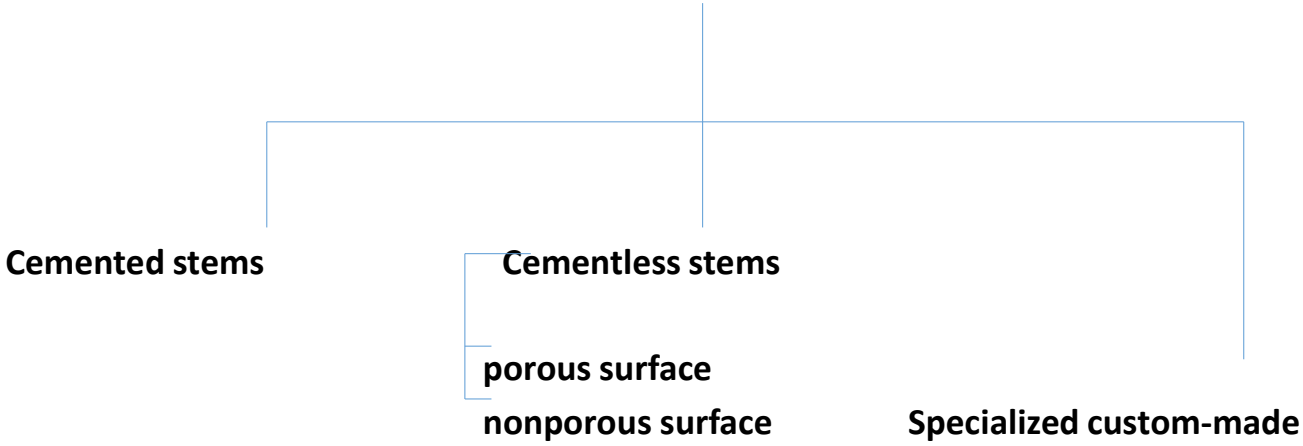
Ti-6Al-4V is chosen over commercially püre (CP) titanium are related to its **superior mechanical properties**, **biocompatibility**, and **performance** in load-bearing implants like hip replacements.

•Ti-6Al-4V (annealed) has a tensile strength of **~830-900 MPa**.
•CP titanium (Grade 2) has a tensile strength of **~300-500 MPa**.

Ti-6Al-4V is easier to work with in advanced manufacturing techniques such as **additive manufacturing (3D printing)**, **casting**, and **machining**.

TYPES OF FEMORAL COMPONENTS

The Femoral Component Replaces the natural femoral head portion or all of the femoral neck and bony elements in the proximal femur between Greater and lesser trochanters.



CEMENTED STEMS

Most designers favour- **cobalt chrome alloy**

PMMA cement is the standard for femoral component fixation

Disadvantages- Debonding, Mechanical loosening, Extensive bone loss with fragmented cement

Earlier the original Charnley's component was about 130 mm long. But current stem design ranges from 120-150mm.

2. Cementless stems with porous surface

Fixation is more biological.

Material- titanium alloy/ Cobalt-Chromium alloy

Bone ingrowth into porous metal surface

Requires: a)immediate mechanical stability at the time of surgery

b) intimate contact between porous surface and viable host bone

So, surgical technique and instrumentation need to be more precise than cemented counterpart

Cementless stem Types

There are 6 types of Cementless femoral components.

Type I – Type V are straight stems and fixation area increases with type.

Type VI is anatomical: posterior bow in metaphysial portion and anterior bow in diaphysial portion corresponding to the geometry of femoral canal.



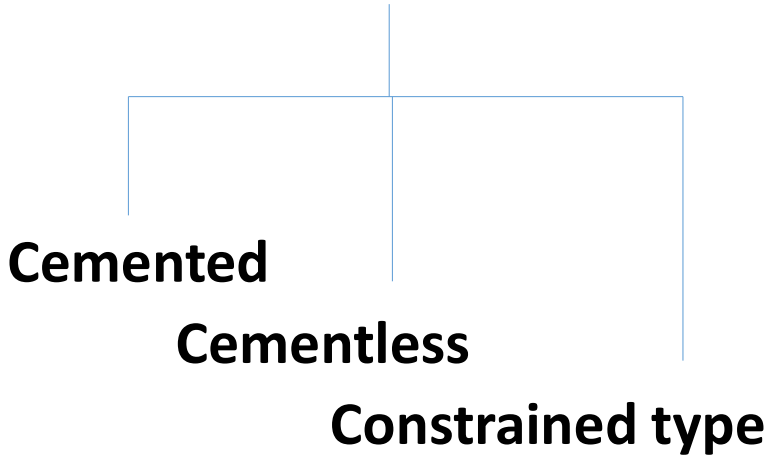
Specialized custom-made

Specialized femoral components for replacement of variable length of proximal femur. Stem can be combined with TKR to replace entire femur



Acetabular component designs & Materials

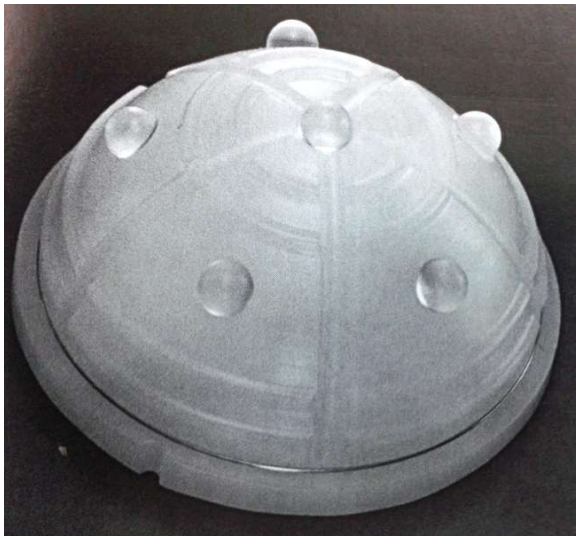
ACETABULAR COMPONENTS



Cemented acetabular component

PMMA spacers (3 mm) are incorporated into polymerizing cement, yeilding uninterrupted cement mantle

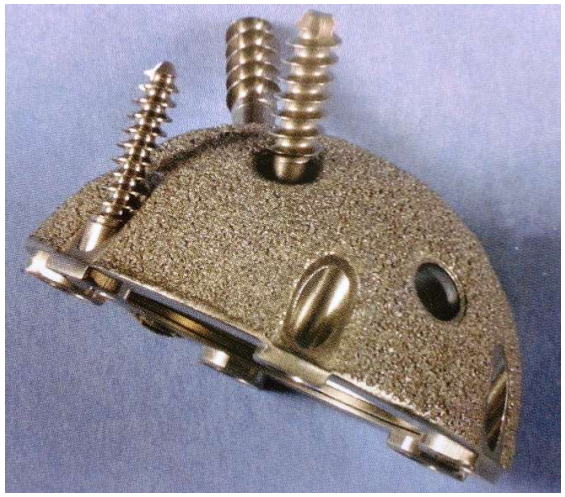
Satisfactory in elderly, low demand patient, Tumour reconstruction, and in revision arthroplasty.

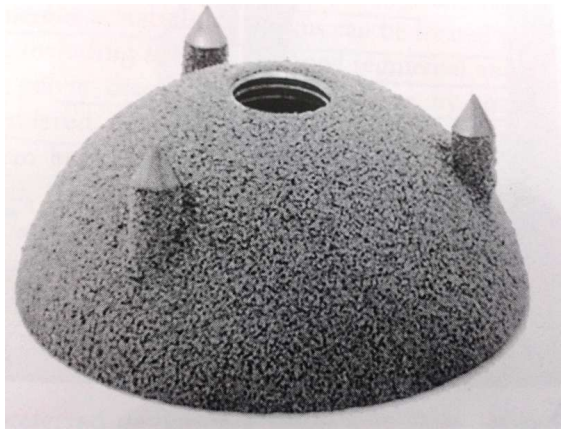


Cementless acetabular components

1. Porous coated for bone-ingrowth

2. Fixation with
trans acetabular screw





Constrained acetabular components

Mechanism to lock the prosthetic femoral head into the polythene liner





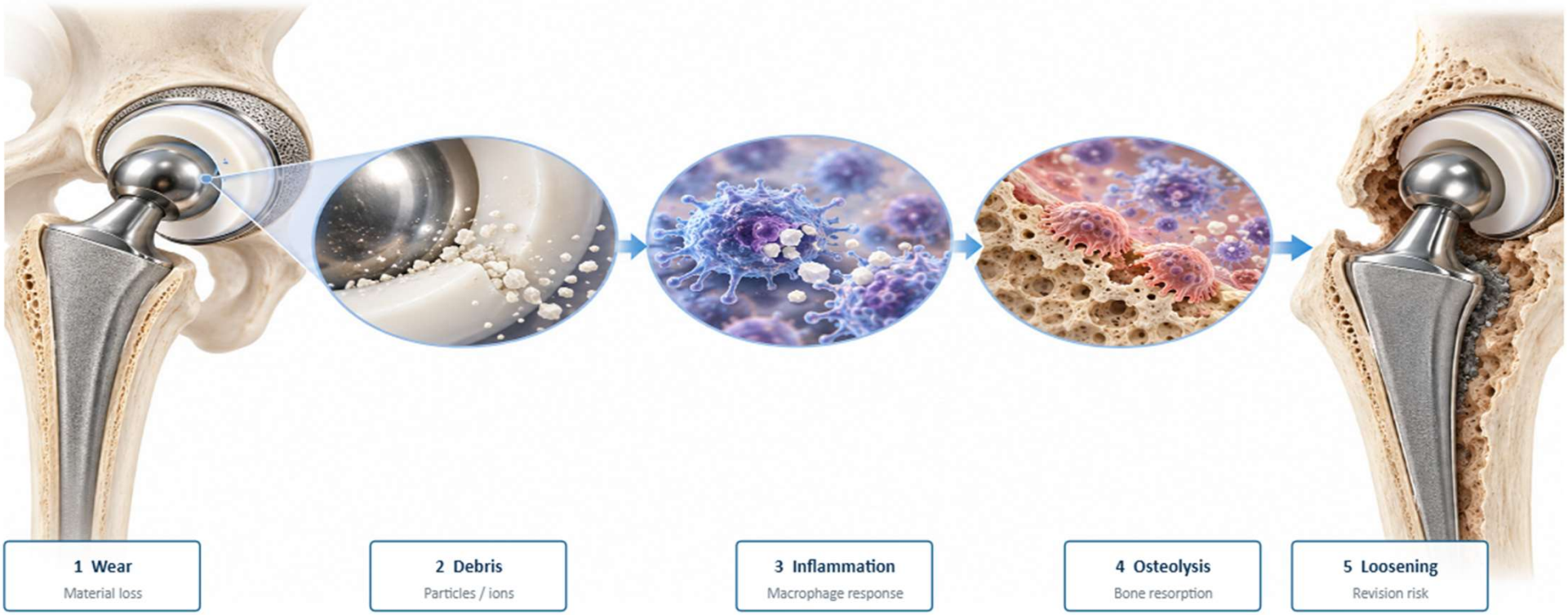


Ideal bearing surface

1. Low coefficient of friction
2. Small volume of wear particle generation
3. Low tissue reaction to wear particles
4. High resistance to third body wear
5. Enough deformation of articular surfaces to permit adequate fluid film lubrication during the stance phase without increasing wear

Wear debris can convert a good bearing into a biological failure

The amount, chemistry and size of debris all matter.



Materials selection for acetabular cup and femoral head

Metal-on-metal implants were first developed in the 1960s but have since been greatly improved. Cobalt chrome is a popular choice, although some studies have found that metal-on-metal implants can cause elevated levels of the metal ions in urine and the bloodstream. This indicates that wear produces particles that enter the body, and which may have an adverse effect. This is particularly a problem for people with poor kidney function.

Studies have shown that increasing the cross-linking in the polyethylene significantly reduces wear, leading to more durable acetabular components, thus increasing the lifetime of an implant

Acetabular liner polymers: the UHMWPE evolution

The target is low wear without sacrificing toughness, fatigue resistance or oxidative stability.

PTFE	UHMWPE	HXLPE	Vitamin-E HXLPE
Benefit	**Benefit**	**Benefit**	**Benefit**
Low friction	Impact toughness Low friction	Much lower wear	Oxidation stabilization Low wear
Challenge	**Challenge**	**Challenge**	**Challenge**
Very high wear Foreign-body response	Wear debris Oxidation	Crosslinking can reduce toughness / fatigue	Processing balance Long-term monitoring

Recent long-term studies continue to support very low wear with modern HXLPE; vitamin-E formulations target oxidative stability.

Metal on metal bearings

Low wear rate

High carbon cobalt chromium alloy

Diametral clearance-gap between the two implants at the equator of articulation.

Smaller clearance produce films for lubrication and reduced wear.

Elevated metal ions in blood that excreted through urine.



Metal on metal bearings

So contraindicated in impending renal failure.

Placental transfer occur of these metal ions.

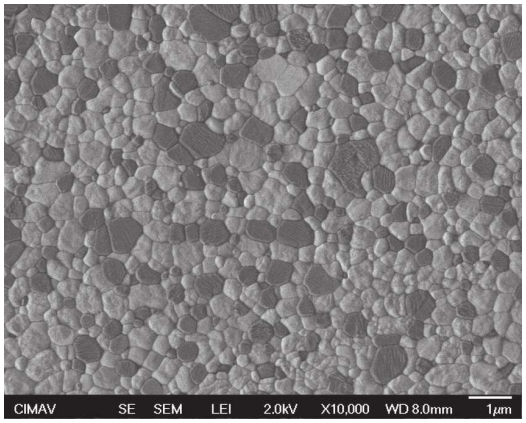
Delayed type hypersensitivity (aseptic lymphocytic vasculitis associated lesions)

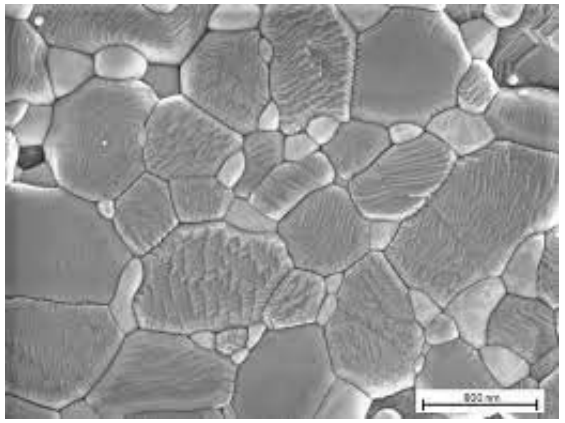
Pseudotumour

Recommendation for symptomatic patients is measurement of blood cobalt and chromium ion level and/MRI or USG.

Materials selection for acetabular cup and femoral head

Ceramic-UHMWPE success:
These improved wear resistance properties are dependent on the small, homogeneous grain size in the ceramic, so the microstructure must be carefully controlled during the manufacturing process.





CERAMIC ON CERAMIC BEARINGS

ALUMINA CERAMIC IS USED.

HIGH DENSITY, HYDROPHILLIC, SMOOTHER THAN METAL.

CERAMIC IS HARDER THAN METAL AND MORE RESISTANT TO SCRATCHING.

LINEAR WEAR RATE IS 4000 TIME LESS THAN COBALT CHROME ALLOY ON POLYETHYLENE.



DISADVANTAGE

CHIPPING/COMPLETE FRACTURES

IMPLANT MALPOSITION

STRIPE WEAR

SQUEAKING

OSTEOLYSIS



Fig. 3 – Fractured ceramic liner of acetabular component.

Materials selection for acetabular cup and femoral head

They rub against each other. So you have to select accordingly.

Each pairing has specific properties related to wear, longevity, and biocompatibility

Metal-Metal:Used in some older hip replacements but less common now due to concerns about metal ion release and wear particles, which can cause **adverse tissue reactions**.

Ceramic-Ceramic:Known for **excellent wear resistance** and biocompatibility. **Ceramic-on-ceramic** pairings reduce the risk of wear particles but more prone to **fractures.**

Metal-UHMWPE (Ultra-High Molecular Weight Polyethylene):The most common combination for hip replacements due to good durability and **low cost**. UHMWPE is a **high-wear-resistant polymer**, but over time, it may generate wear debris that can lead to osteolysis.

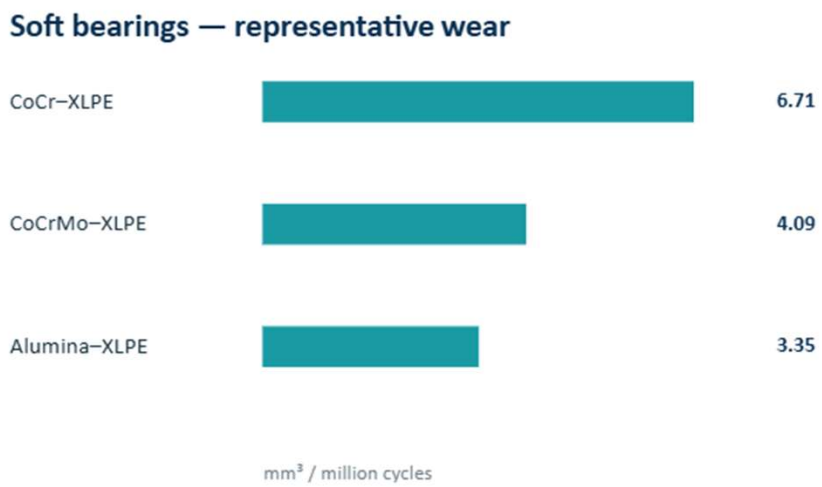
Ceramic (Alumina-Zirconia) - UHMWPE:This combination is considered one of the **best solutions for wear resistance** because **ceramic femoral heads** paired with **UHMWPE acetabular cups** reduce wear rates significantly compared to metal-on-polyethylene.

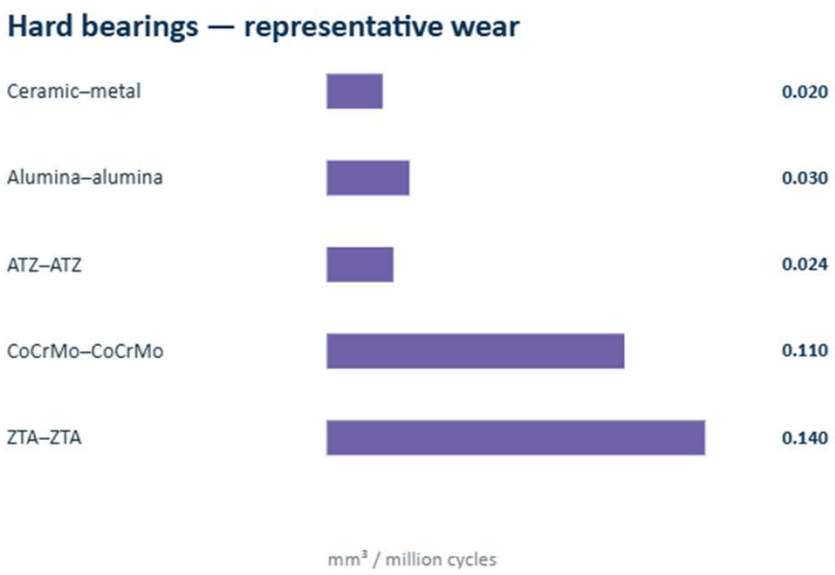
Comparison of Materials used in Total Hip Arthroplasty

Prosthesis	Advantages	Disadvantages
Metal-on-polyethylene	Large volume of evidence to support use Predictable lifespan Cost effective	Polyethylene debris leading to aseptic loosening
Metal-on-metal	Potentially longer lifespan than polyethylene due to reduced wear Larger femoral head - therefore lower dislocation rate	Metallosis Potential carcinogenic effect of metal ions
Ceramic-on-ceramic	Low friction Low debris particles Inert substance	Expensive Require expert insertion to prevent early damage Can produce noise on movement

Bearing couples: every pairing solves one problem and creates another

Representative simulator wear data from the review show the scale difference between soft and hard bearings.





Interpret carefully: simulator protocols and material generations differ. The trend—not exact ranking—is the teaching point.

Materials selection for acetabular cup and femoral head

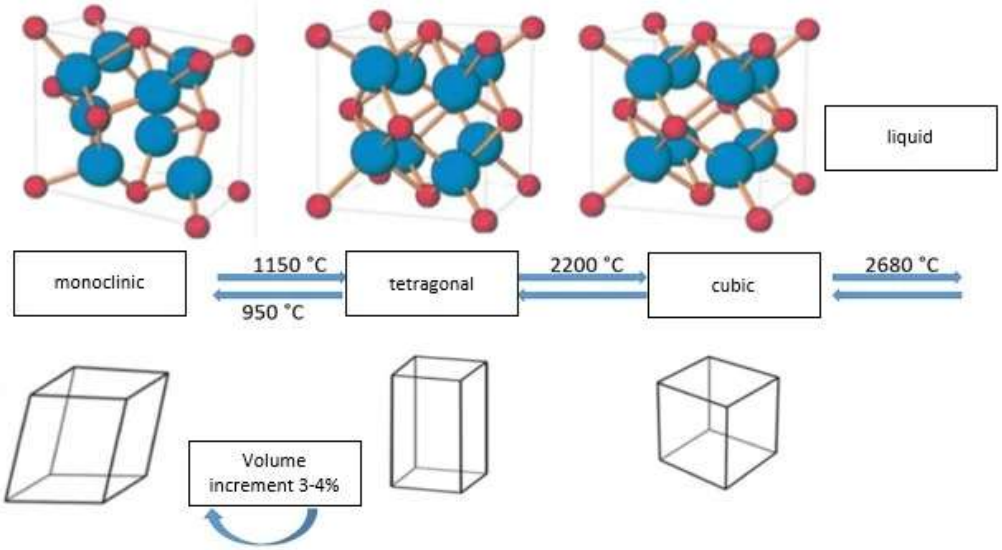
How to impede crack formation and failure in ceramics?

•**Zirconia-Toughened Alumina (ZTA)**: One of the most successful replacements for zirconia is ZTA, which is a composite material that combines alumina (Al_2O_3) with small amounts of zirconia. ZTA benefits from the high wear resistance of alumina and the increased toughness provided by zirconia. This material has better long-term stability and lower risks of phase transformation compared to pure zirconia.

•**Alumina Ceramics**: Pure **alumina** (Al_2O_3) ceramics have long been used in femoral heads due to their excellent wear resistance and bioinertness. Modern alumina is often preferred over zirconia because it avoids the phase transformation issues seen in zirconia.

Transformation toughening in ZTA

Zirconia is a polymorphic material



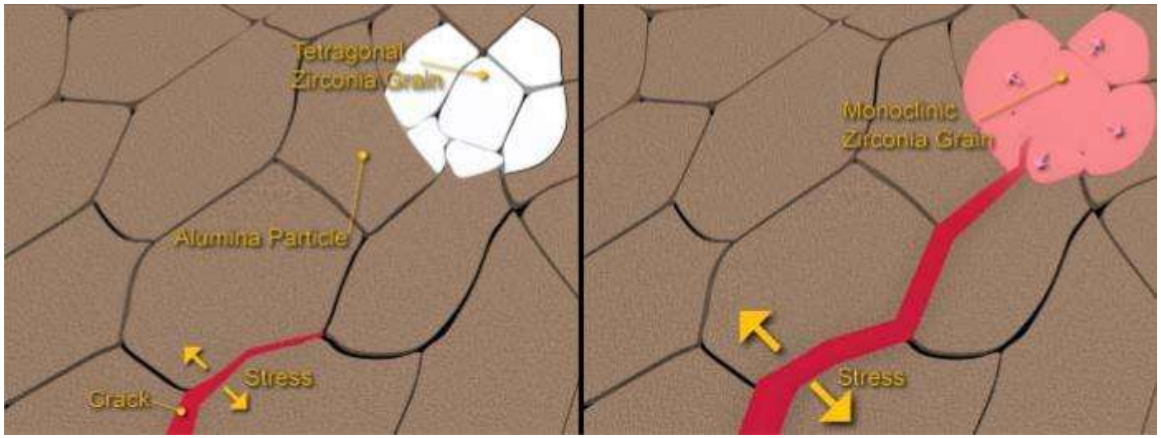
The cubic and tetragonal phases of zirconia can be increased by adding dopants like **Y_2O_3 (yttria)**.

- **7 mol% Y_2O_3**: produces **cubic zirconia**
- **2-6 mol% Y_2O_3**: results in **partially stabilized zirconia (PSZ)**

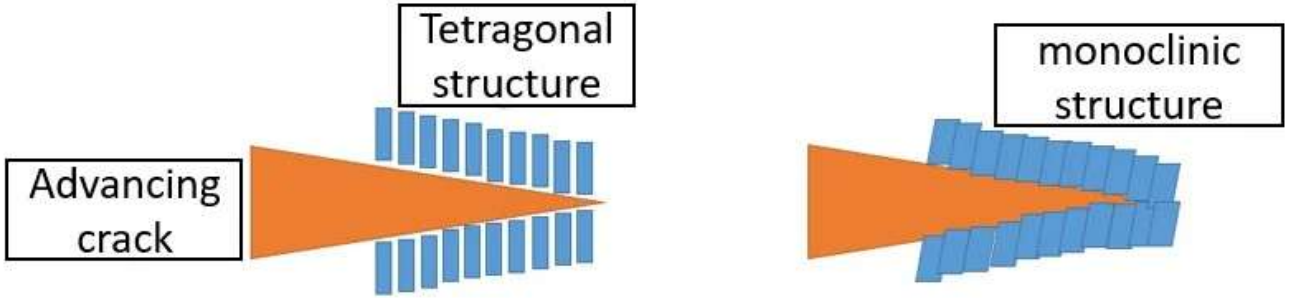
Approximately **10% of the tetragonal phase** is present within a cubic matrix.

phase transformation (tetragonal to monoclinic) helps improve fracture toughness

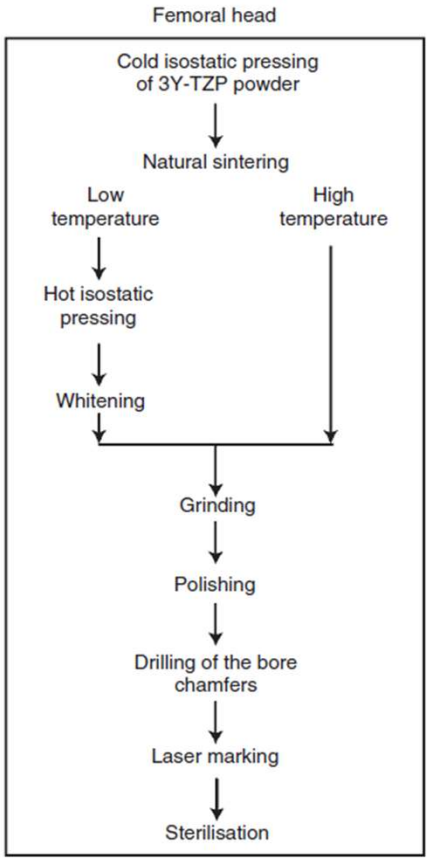
Transformation toughening in ZTA



Example of phase transformation and crack propagation (Kyocera Medical).



Production method



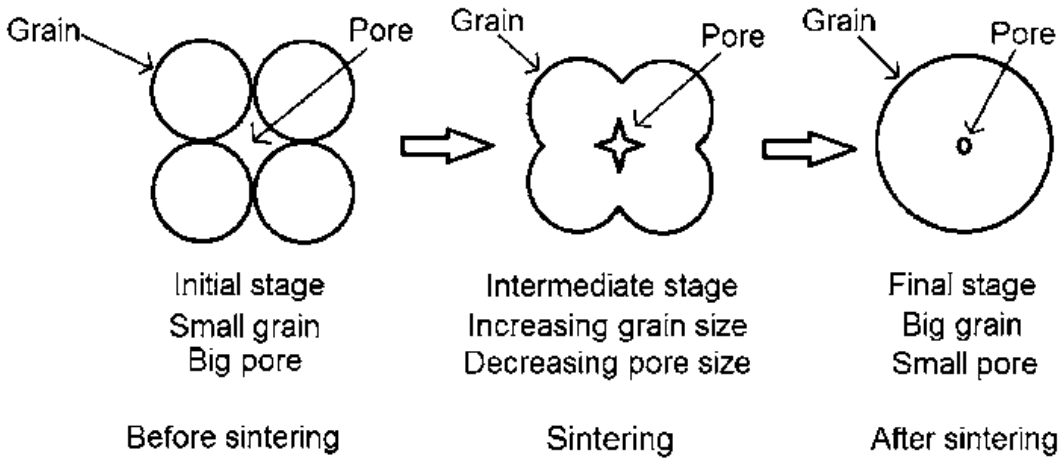
11.2 Synopsis of the process of zirconia femoral heads.

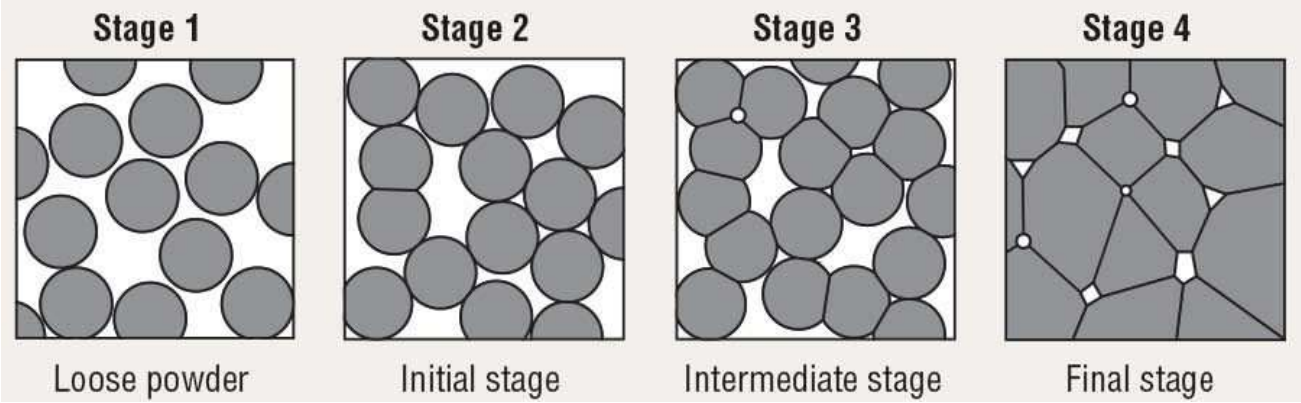


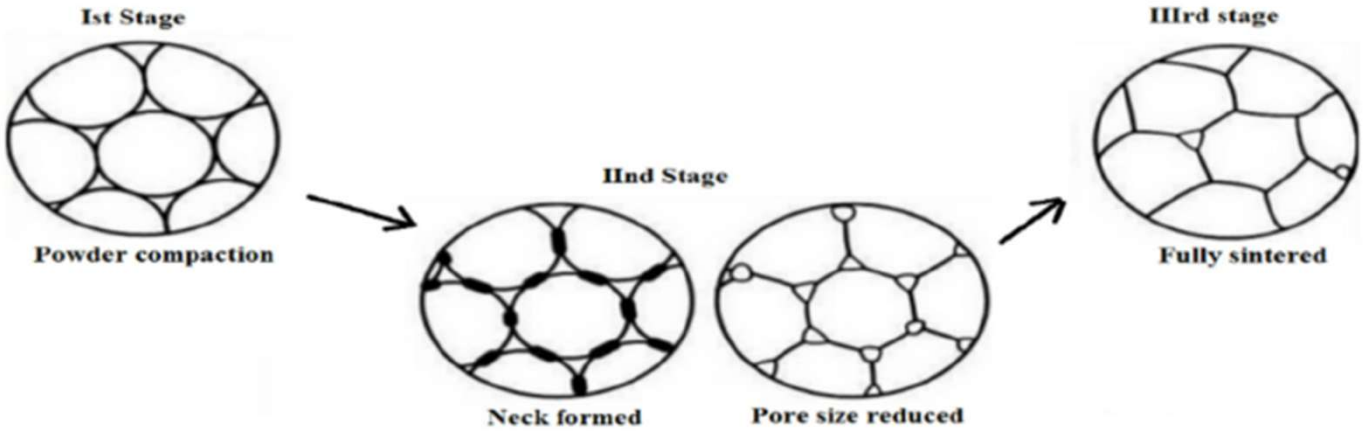
SINTERING

Sintering
- *Controlled Atmospheres*
- *Sinter/Hardening*
- *Vacuum Sintering*
- *Dimensional Changes*
- *Activated Sintering*
- *Liquid Phase Sintering*

$T_{sintering}$: 85 % of $T_{melting}$




Stage 1
Stage 2
Stage 3
Stage 4
Loose powder
Initial stage
Intermediate stage
Final stage


Ist Stage
Powder compaction
IInd Stage
Neck formed
Pore size reduced
IIIrd stage
Fully sintered

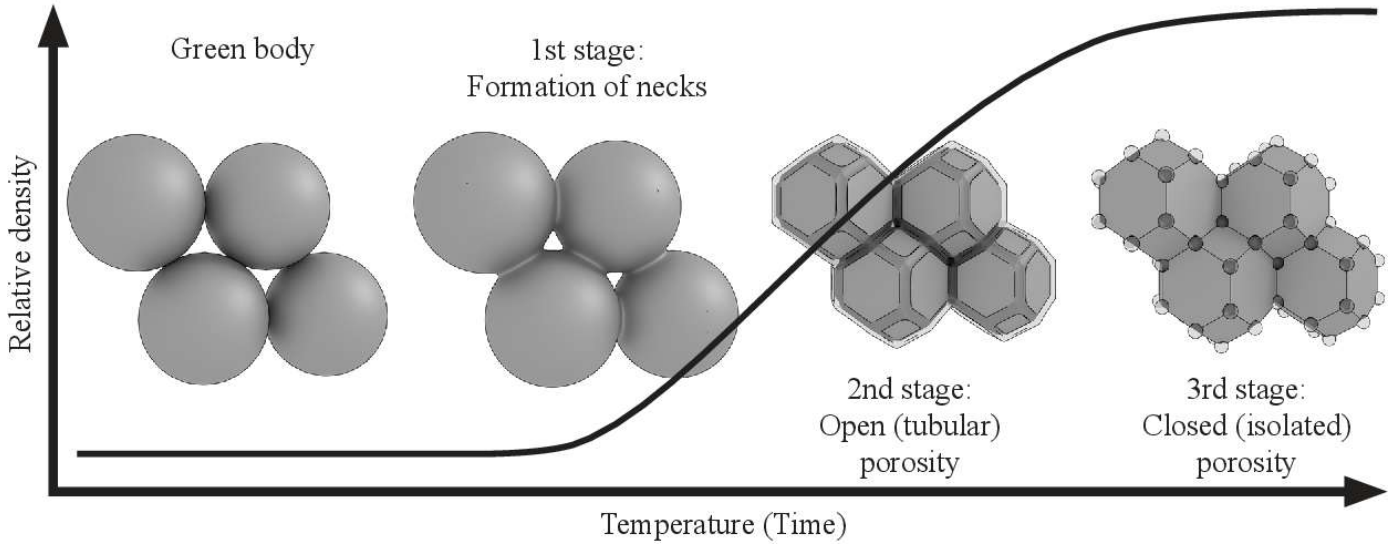


Figure 6: Stages of sintering.

The relationship between strength and grain size in ceramics

•**Less Surface Roughness**: Smaller grains can result in a smoother surface, which helps reduce wear. Smoother surfaces create less friction and minimize the chances of abrasive effects causing wear.
•**Higher Hardness**: Ceramic materials are known for their high hardness, and smaller grains often lead to increased hardness. Greater hardness can improve the material's resistance to wear, as it becomes less prone to deformation or abrasion.
•**Improved Durability**: While ceramics are known for their hardness, they can also be brittle. Smaller grains can enhance the durability or resistance to fracture of ceramic materials, making them more reliable in demanding applications.
•**Reduced Crack Propagation**: Smaller grains can help prevent the spread of cracks or defects within the material. This is crucial for avoiding sudden failure and maintaining wear resistance over time.
•**Homogeneity**: Smaller grains contribute to a more homogeneous microstructure, which can result in consistent wear properties throughout the material.

Failure modes mapped to materials

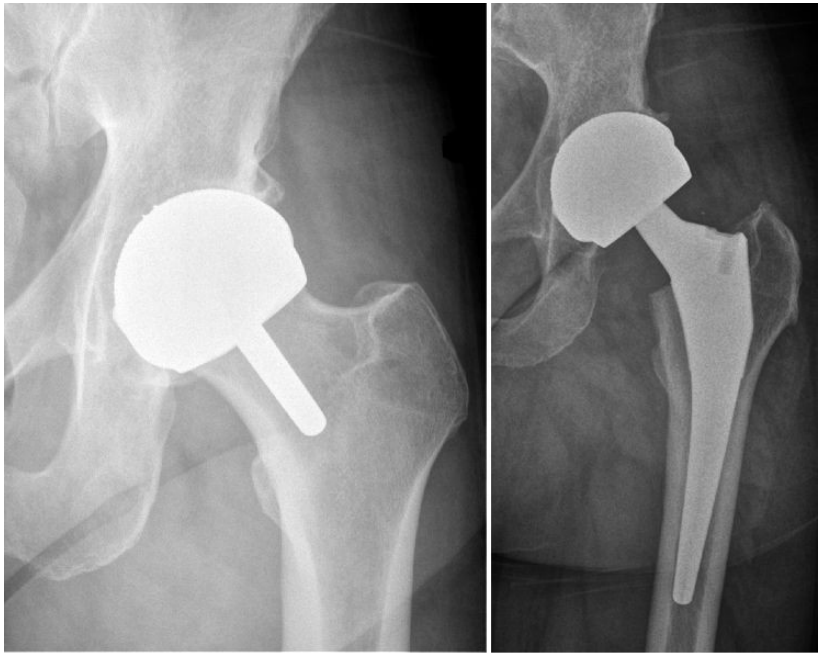
A hip prosthesis fails when a material limitation meets a demanding local condition.

Failure mode	Materials trigger	Materials response
Aseptic loosening	PE wear debris / poor fixation	HXLPE • porous Ti/Ta • better surfaces
Stress shielding	High stem stiffness	Ti / β-Ti • shorter stems • porous design
Corrosion / ions	CoCr modular junctions / MoM	Reduce metal–metal contact • surface control
Ceramic fracture	Brittleness + assembly defects	ZTA • cleaner taper • better processing
Liner damage	Oxidation / fatigue / thin section	HXLPE • Vit-E HXLPE • geometry control
Edge loading	Cup position + clearance + hard bearings	Design + surgical positioning + robust materials

Materials choice reduces risk; it cannot eliminate geometry, surgical and biological causes.

Alternatives & future direction

Other option for hip replacement: hip resurfacing



Hip Resurfacing

Traditional Total Hip Replacement

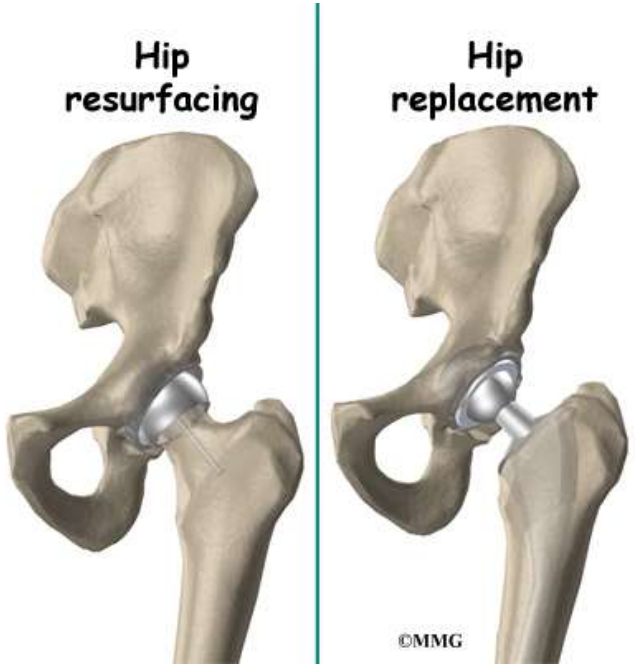
In a traditional total hip replacement surgery, both the head of the thigh bone (femoral head) and the damaged socket (acetabulum) are removed and replaced with components made of metal, plastic, or ceramic.

In **hip resurfacing**, however, the femoral head is not removed; instead, it is covered with a smooth metal cap. The damaged bone and cartilage inside the socket are removed, and they are replaced with a metal shell, similar to that used in traditional total hip replacement surgery.

Metal-on-metal hip resurfacing often uses larger femoral heads. A larger head size can improve joint stability, reduce the risk of dislocation, and enhance joint function.

Hip resurfacing





Future Advances in THR

Minimally invasive surgery :

Gaining popularity in recent years, minimally invasive techniques are currently being developed. The use of a single-incision, less than 10 cm in length using conventional surgical approaches, provides soft-tissue sparing and bone conservation options.



Computer-assisted surgery :

Computer-assisted total hip replacement utilizes digital image systems to map the position of surgical instruments in relation to anatomical landmarks, helping to obtain reproducible and accurate placement of implants. Computer navigation may improve the accuracy of prosthesis positioning but, despite its obvious advantage with respects to reducing asymmetric wear, this has not yet been shown to have a clinical benefit.



Why a hip prosthesis is a materials problem

The implant must reproduce a joint while surviving millions of load-and-slide events in a corrosive biological environment.

SYSTEM VIEW

1 **Carry load**	2 **Move smoothly**	3 **Stay fixed**	4 **Stay biologically quiet**
Body weight + muscle forces Static + cyclic loading	Low friction Stable lubrication	Bone–implant load transfer Long-term fixation	Low toxicity Minimal harmful debris

The “best material” depends on the component role.

STEM → NECK → HEAD → LINER → CUP/SHELL

Different loads • different interfaces • different failure modes

What would a “perfect” hip prosthesis require?

No single material satisfies every requirement; success comes from matching materials to roles.

1 Biocompatibility

No toxic or inflammatory response

2 Strength + fatigue

Survive repeated high loads

3 Stiffness match

Limit stress shielding

4 Low friction

Smooth articulation

5 Very low wear

Minimal debris generation

6 Corrosion resistance

Stable in body fluids

7 Toughness

Resist crack/fracture

8 Fixation

Immediate + biological stability

9 Manufacturability

Reliable processing + sterilization

Ideal target: function for decades with minimal revision risk

Design takeaway: the “perfect hip” is a system, not a material

Component-specific optimization is the central materials lesson.

A strong current materials architecture

Ti-alloy stem: fatigue + fixation + lower stiffness
Porous Ti / Ta shell: biological fixation
Ceramic head: hard, smooth, low-wear counterface
HXLPE liner: low friction + very low modern wear
Engineered surfaces: osseointegration without sacrificing bulk strength

Final principle **Match each material to its local job — then engineer the interfaces.**



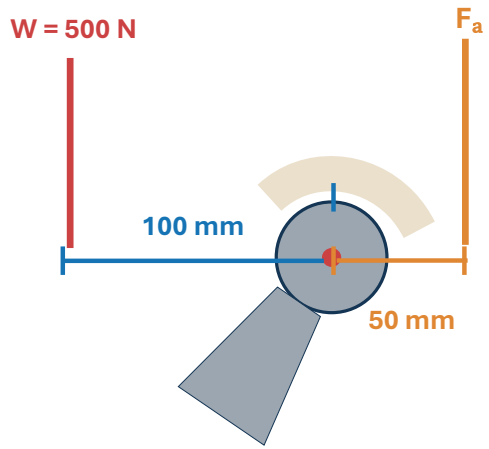
Problem 1 — Single-leg stance

QUESTION

During single-leg stance, an effective body-weight force of 500 N acts 100 mm medial to the hip-joint center. The hip abductors act with a 50 mm moment arm.

Assume static equilibrium and vertical forces.

Calculate:
1. The required abductor muscle force.
2. The approximate hip joint reaction force.



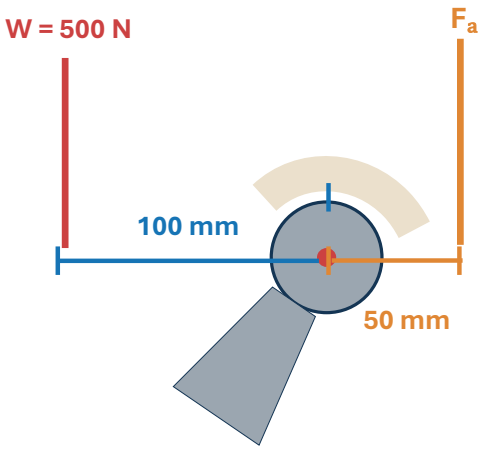
Problem 1 — Single-leg stance

SOLUTION

Moment balance about the hip center:
$500 \times 100 = F_a \times 50$

$F_a = 1000$ N

Vertical force balance:
$R = W + F_a$
$R = 500 + 1000 = 1500$ N



Abductor force = 1000 N

Joint reaction = 1500 N = 3.0× W

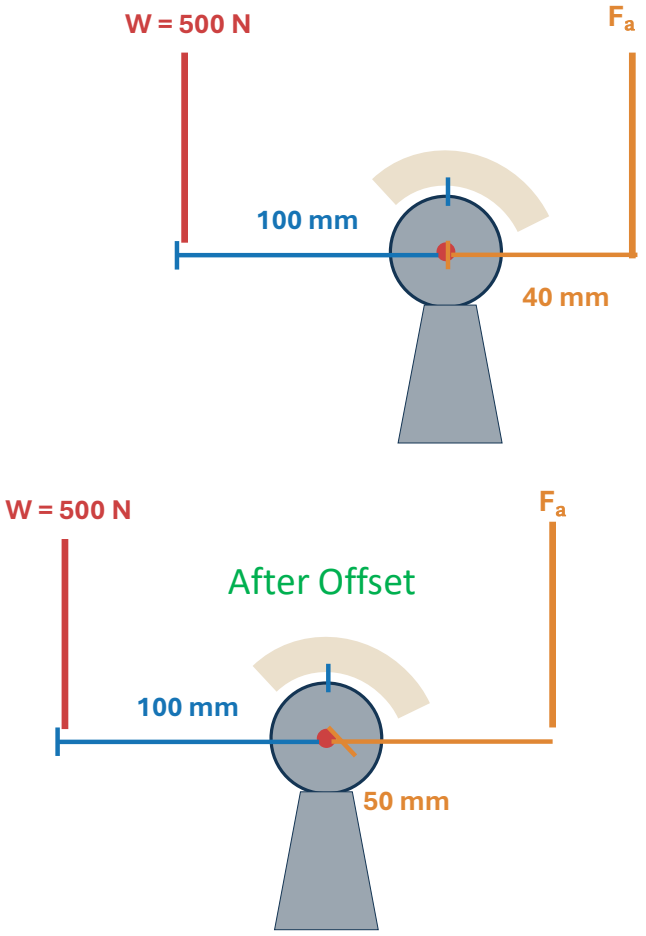
Problem 2 — Restoring femoral offset

QUESTION

A patient has an effective body-weight force of 500 N acting 100 mm medial to the hip center.

Before reconstruction, the abductor moment arm is only 40 mm. After total hip replacement, femoral offset is restored and the abductor moment arm becomes 50 mm.

Calculate the abductor force and joint reaction force before and after reconstruction, and the percentage reduction in joint reaction force.



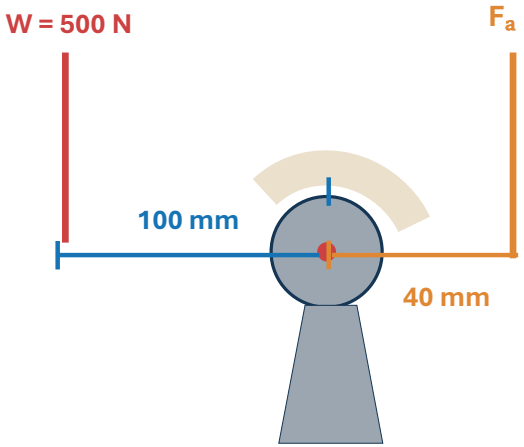
Problem 2 — Restoring femoral offset

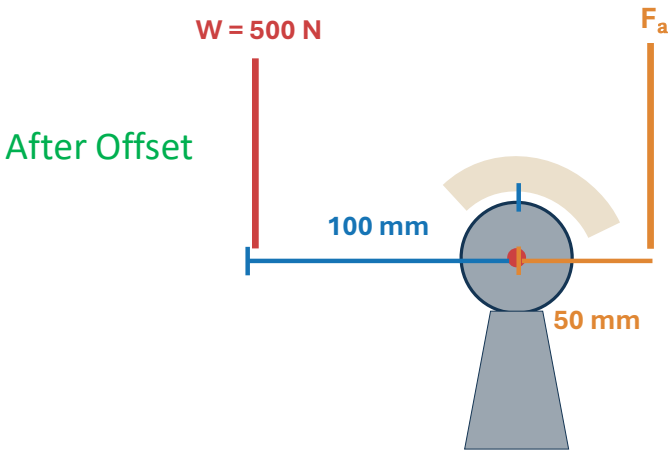
SOLUTION

Before THR:
500 × 100 = F_a × 40 → F_a = 1250 N
R = 500 + 1250 = 1750 N

After offset restoration:
500 × 100 = F_a × 50 → F_a = 1000 N
R = 500 + 1000 = 1500 N

Reduction = (1750 − 1500) / 1750 × 100 = 14.3%





Before: R = 1750 N

After: R = 1500 N

Reduction = 14.3%

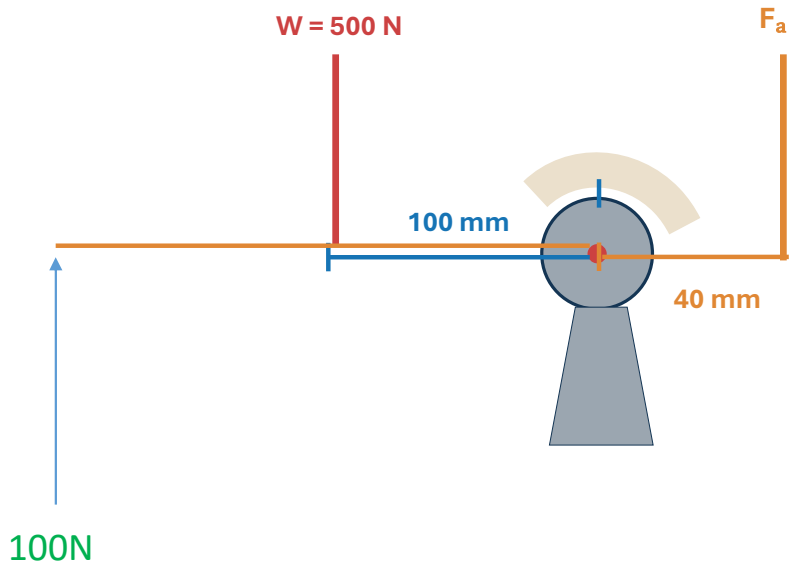
Problem 3 — Effect of a contralateral cane

QUESTION

A patient uses a cane in the hand opposite the painful hip during single-leg stance. The effective body-weight force is 550 N at a 100 mm moment arm. The abductor moment arm is 50 mm.

The cane provides an upward force of 100 N at a point 300 mm from the hip center.

Calculate:
1. The required abductor force.
2. The joint reaction force.
3. The percentage reduction in joint reaction compared with no cane.



Problem 4 — Weight gain + loss of offset

QUESTION

Consider a simplified single-leg stance.

Baseline: effective body-weight force = 600 N, body-weight moment arm = 100 mm, abductor moment arm = 50 mm.

Later, body weight increases so the effective force becomes 700 N, while loss of femoral offset reduces the abductor moment arm to 40 mm.

Calculate the joint reaction force in both cases and the percentage increase.

Problem 5 — Vector joint reaction force

QUESTION

During single-leg stance, an effective body-weight force of 600 N acts 100 mm medial to the hip center. The abductor muscle has a perpendicular moment arm of 50 mm and its line of action is 20° from vertical.

Assume the abductor force acts downward and laterally on the pelvis.

Calculate:
1. The abductor force from moment equilibrium.
2. Its vertical and horizontal components.
3. The magnitude and direction of the resultant hip joint reaction force.